Workflows

Every Job the Workbench Does, Read by a Council of Researchers

HARISH GOVARDHAN DAMODAR

First edition — October 2026

A supplement to *Agentic Design Patterns & Guardrails*: every workflow in the workbench — catalog, planning, labs, counsel and knowledge — read by a council of researchers, with every scenario assigned to the workflows that serve it and a playbook for all 26 catalog domains. Generated from the catalog and the app; nothing in it is hand-typed from memory.

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Preface

This supplement is the manual the other books assume. Patterns tells you what to build, Rogue Agents tells you what to fear, OpenShell tells you where to stand — this book tells you which tab to open, in which order, and what done looks like. Every workflow below is read the same way: where to start, what to do, and what leaves the tab as evidence.

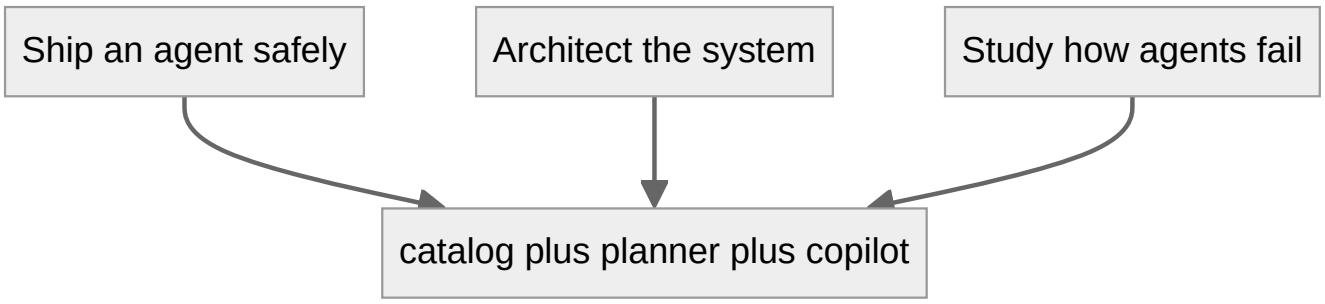
Three questions organize it around the three jobs on the Overview tab: plan a deployment safely, architect the system, and study how agents fail. The catalog workflows feed the planning workflows, the labs rehearse what the catalog teaches, and the copilot, design docs and bookshelf hold the whole program together.

This book was written and compiled from study and experiment observation notes of Harish Govardhan Damodar and generated by the *AI Agent Patterns & Guardrails* catalog by a deterministic script (`tools/build_workflows_book.py`), then typeset to PDF with headless Chrome and the vendored Mermaid engine (`tools/render_pdf.mjs`) — the same Mermaid that renders the diagrams in the app, so no diagram in this book was drawn by hand or screenshotted. All experiments performed in local sandbox build and maintained by Harish Govardhan Damodar and his agents, work extensively uses local open weight model for experimentation and intelligence serving, for a live demo / suggestion feel free to drop a DM in my linkedin page (https://www.linkedin.com/in/harishgovardhan-d-44b22b47/). All records, suggestions advice is from personal experiments ran on my personal setup only and does not represent the organisation I work for.

Council deliberation.

- **Continual-learning researcher:** I frame this supplement as the manual the other books assume: patterns tell you what to build, rogue drills tell you what to fear, OpenShell tells you where to stand — this book tells you which tab to open, in which order, and what done looks like. Every workflow below is read the same way: where to start, what to do, what leaves the tab as evidence. My dispute is with readers who treat the app as a dashboard to glance at; it is a workbench to run jobs on. Monday morning, pick one job and walk it end to end instead of browsing.
- **Memory-systems researcher:** I read the workbench as institutional memory made clickable: projects, libraries, saved runs, quarantine boards and promoted drafts persist in the browser so an investigation survives the week it spans. The workflows matter less than what they keep — the export, the backlog, the report that travels to the next planning cycle. Others emphasize speed through a tab; I insist the artifact at the end is the point. Monday morning, finish one workflow with an export rather than a screenshot.
- **Evaluation methodologist:** I present the preface as a contract about provenance, matching the other supplements. Every count comes from the catalog at build time, every action name is validated against the shipped scripts, every scenario-to-workflow assignment follows a stated rule. The book fails its build rather than drift from the app it describes. My caution is against quoting workflows from memory after the app moves on; regenerate and re-check. Monday morning, verify one action name in the tab before citing this book.
- **Safety researcher:** I treat this book as the operating discipline the catalog deserves: a swarm nobody rehearses with is a demo, a blueprint nobody stress-tests is a sketch, a gap nobody backlogs is a surprise with a date. Workflows turn knowledge into practice on a schedule. My disagreement is with teams that read the catalog and change nothing; reading without a workflow is entertainment. Monday morning, put one workflow on the team calendar — bench quarterly, quarantine drills monthly, gap review per release.

Introduction: three jobs, one workbench



Three jobs, one catalog.

The catalog behind this book holds 107 patterns, 54 failure scenarios and 191 cascades across 26 domains, served through 15 tabs in six sections plus the command palette. Catalog edition: counts below are read from data/*.json at build time. Every scenario below is assigned to the workflows that serve it by a stated rule; every domain chapter lists its real catalog scenarios.

The assignment rule, printed here so it can be argued with: every scenario starts in Scenarios, gets rehearsed in Rogue ops against its archetype, and is read against its domain gap in Coverage. Scenarios with mock runs add Live traces; boundary shapes (hijack, exfiltration) add the Guardrail lab; authority shapes (abuse, collusion, spawning) add Blueprints; high-stakes domains add the Deployment planner. The copilot and design docs serve every scenario and get their own chapters instead of per-row mentions.

Council deliberation.

- **Continual-learning researcher:** I define the workbench as three jobs sharing one catalog: plan a deployment safely, architect the system, and study how agents fail. Every tab serves at least one job, and the palette jumps between them without losing context. My dispute is with tab-tour onboarding that shows everything and assigns nothing; start from the job and the tabs arrange themselves. Monday morning, name your job before opening the app.
- **Memory-systems researcher:** I emphasize that all memory here is local: projects, libraries, runs, boards and drafts live in the browser store on your machine, exportable as JSON or Markdown, working from a static host with no backend at all. Nothing you rehearse has to leave the room. Others focus on features; I insist ownership of the working record is the feature. Monday morning, export one artifact and confirm you can read it without the app.
- **Evaluation methodologist:** I read the introduction as scoping what this book proves versus what it shows. What it proves: the inventory — every tab, every action name validated against the shipped code, every scenario assigned to workflows by a stated rule, every domain covered. What it shows: walkthroughs of each workflow in its canonical order. The honest line between the two runs through the whole book. Monday morning, restate that boundary whenever a walkthrough travels upward as a promise.
- **Safety researcher:** I translate the workbench into an operating stance: plan before granting, rehearse before trusting, review before releasing, and write down who owns each step. The workflows below are the standing orders; the council readings are the debate behind them. My disagreement is with teams that improvise incident response from a catalog; rehearse the handoffs until they are boring. Monday morning, verify the halt works, the board is reachable, and the backlog has an owner.

Part I — The workbench and its three jobs

Chapter 1: The three jobs

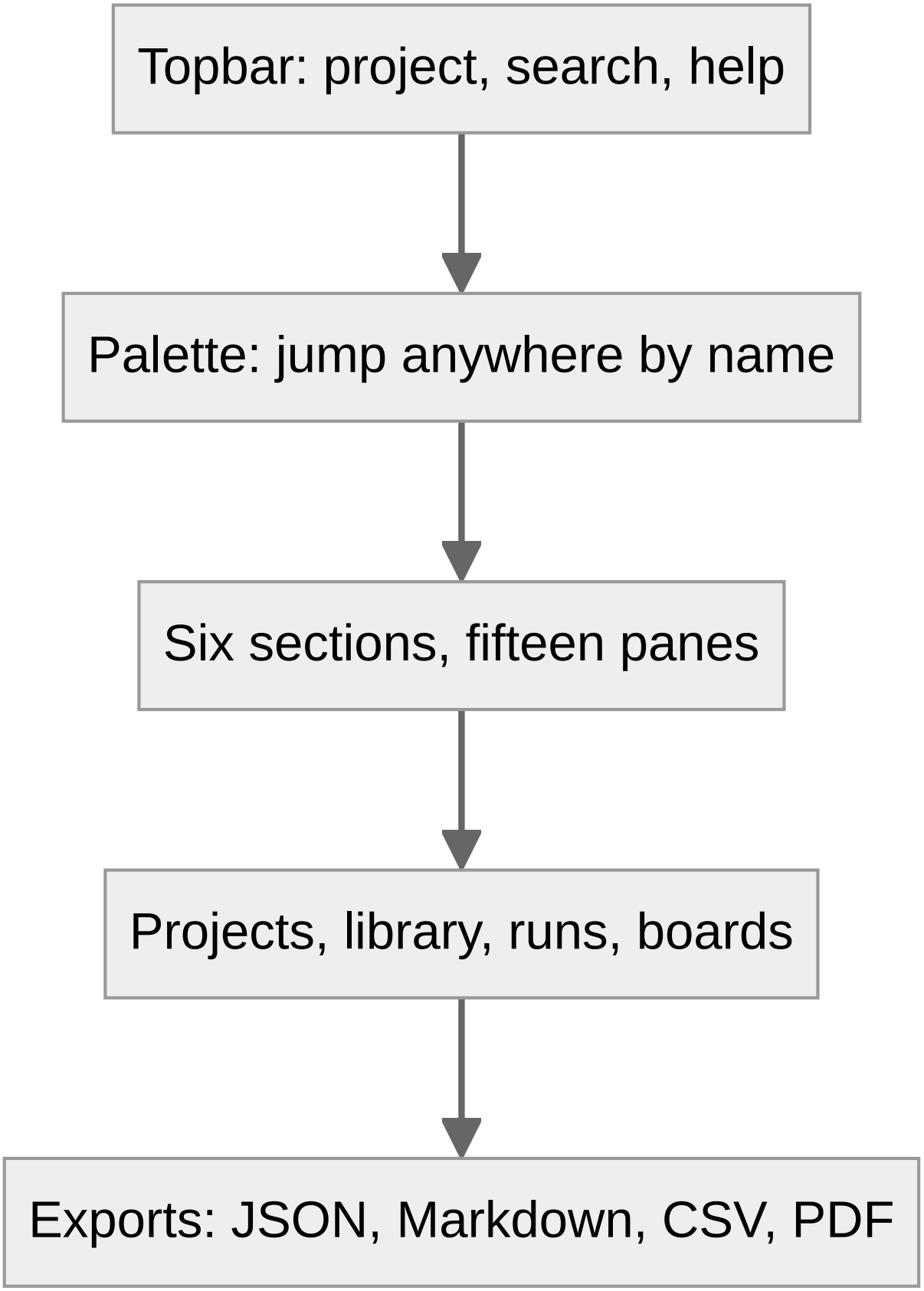
Ship an agent safely, architect the system, study how agents fail. Each job lines up a different slice of the catalog, the planner and the copilot behind it, so the first click already has a purpose — and each job ends with an artifact, not a browse history: a readiness score with a rollout plan and an ADR, stages with controls and estimates in the library, or rehearsed shapes with measured bands.

Council deliberation.

- **Continual-learning researcher:** I read the three jobs as the only onboarding that sticks: ship an agent safely, architect the system, study how agents fail. Each job lines up a different slice of the catalog, the planner and the copilot behind it, so the first click already has a purpose. My dispute is with feature tours that end at awareness; a job ends at an artifact. Monday morning, enter through one job card and finish what it starts.
- **Memory-systems researcher:** I focus on what each job keeps: the deployment job keeps a readiness score, a stack, a rollout plan and an ADR; the architect job keeps stages, controls and estimates in the library; the study job keeps rehearsed shapes and measured bands. Jobs are distinguished by their artifacts, not their clicks. Others emphasize the starting point; I insist the kept record is what makes it a job. Monday morning, name the artifact before starting the job.
- **Evaluation methodologist:** I value that the three jobs are testable claims: a deployment plan either scores readiness or it does not, a blueprint either estimates or it does not, a failure study either rehearses or it does not. Vague jobs produce vague outcomes. My caution is against treating the onboarding cards as marketing; they are routers with acceptance criteria. Monday morning, define done for the job you pick.
- **Safety researcher:** I staff the three jobs as the minimum viable program: every team shipping agents should be able to show a deployment plan, a system architecture, and a recent failure rehearsal on demand. If any of the three is missing, that is the gap to work first. My disagreement is with programs that do two of three and call it coverage. Monday morning, check which of the three you cannot produce this week.

Chapter 2: Shell, palette, and projects

Fifteen panes in six sections, one palette that jumps anywhere by name, keyboard shortcuts for the pointerless, and deep links for everything with an identity — drawers, cases, scenarios, runs. The shell is navigable at speed and linkable for handoff: send the link, not the screenshot.



Shell, palette, tabs, memory.

Under the shell sits the memory: the current-project chip, per-tab save stamps, the blueprint library, saved runs and boards and promoted drafts — persisted in the browser, exportable as JSON, working from a static host with no backend at all. Context that survives is context that compounds.

Council deliberation.

- **Continual-learning researcher:** I see the shell as speed without loss of place: the command palette jumps anywhere by name, section tabs keep fifteen panes in six groups, and keyboard shortcuts move without a pointer. Fluency here compounds — every jump you memorize is seconds back forever. My dispute is with mouse-only operators; the palette is the fastest way to work this catalog. Monday morning, learn three palette queries by heart.
- **Memory-systems researcher:** I emphasize the project layer that wraps the shell: current-project chip, per-tab save stamps, library persistence and JSON export mean the workbench remembers across sessions and machines. Context that survives is context that compounds. Others focus on navigation; I insist the shell is a memory system with a search box. Monday morning, export one project and confirm the round trip.
- **Evaluation methodologist:** I treat shell fluency as measurable: palette results ranked, shortcuts discoverable through the help overlay, tab state restorable from the address bar. The deep-link habit matters most — every drawer, case and scenario has an address you can send. My caution is link rot across versions; pin the version with the link. Monday morning, send one deep link instead of a screenshot.
- **Safety researcher:** I map the shell to shared situational awareness: the same palette, the same tabs, the same links mean the on-call and the auditor look at the same objects. Consistency of address is a safety property when minutes matter. My disagreement is with private workflows that cannot be linked or exported. Monday morning, verify your current investigation is shareable as links and exports.

Chapter 3: Yours: offline-first and exportable

Static hosting, browser persistence, deterministic fallbacks when backends are unreachable, and exports that stand alone: JSON for state, Markdown for reports, CSV for backlogs, PDF for books. Nothing rehearsed here needs production credentials or real customer data — mock and open data exercise the same paths.

Council deliberation.

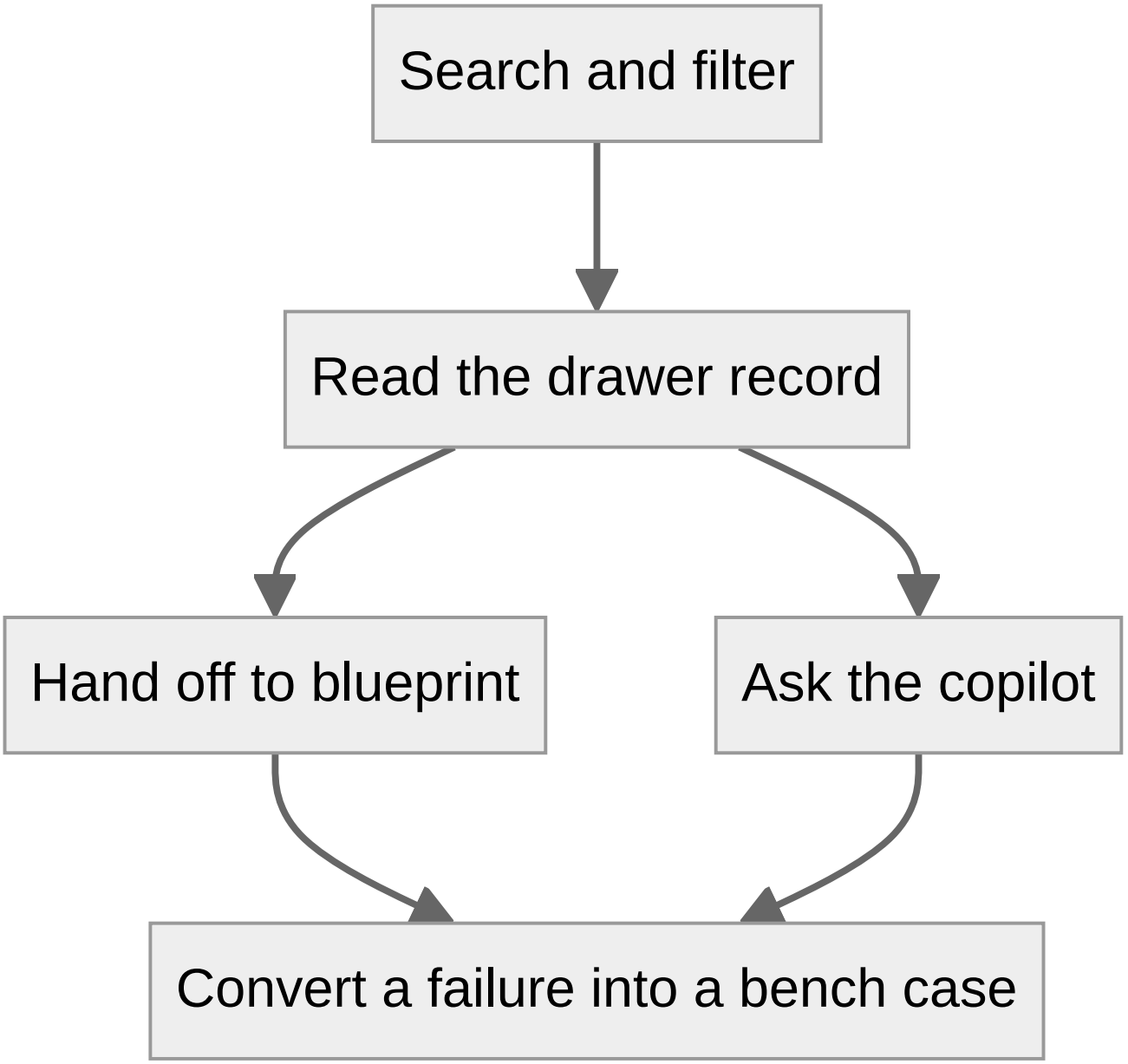
- **Continual-learning researcher:** I read local-first as the precondition for honest rehearsal: static hosting, browser persistence, deterministic fallbacks when backends are unreachable, and exports that stand alone. You can run the whole program on your own machines with your own data. My dispute is with cloud-mandatory tooling for sensitive work; the lab should fit inside your boundary. Monday morning, confirm what leaves your machine when you use each tab — and prefer the answer nothing.
- **Memory-systems researcher:** I focus on the export discipline: JSON for state, Markdown for reports, CSV for backlogs, PDF for books — every artifact readable without the app that made it. Vendor-neutral artifacts outlive every tool. Others celebrate features; I celebrate file formats. Monday morning, open one export in a plain editor and confirm it stands alone.
- **Evaluation methodologist:** I value determinism as the audit property: seeded runs, versioned data files, and builds that fail rather than drift. Reproducibility is what separates rehearsal from storytelling, and local-first makes it cheap. My caution is stale caches — the shell checks its own freshness and says so. Monday morning, confirm your outputs carry versions and seeds.

- **Safety researcher:** I staff local-first as a containment boundary for the work itself: rehearsals of exfiltration, rogue behavior and live fire should never need production credentials or real customer data. Mock and open data exercise the same paths. My disagreement is with labs that rehearse on production; the fence starts at the lab door. Monday morning, verify no rehearsal touches a production system.

Part II — Catalog workflows

Chapter 4: Patterns: search, filter, drawer, handoff

Search by name, failure, or domain; filter by category; open the drawer for the full record — control flow, failures, protections, links — then hand it off: add it to a blueprint or ask the copilot about it. The handoff is the workflow; a pattern that never leaves the drawer never ships.



Catalog loop.

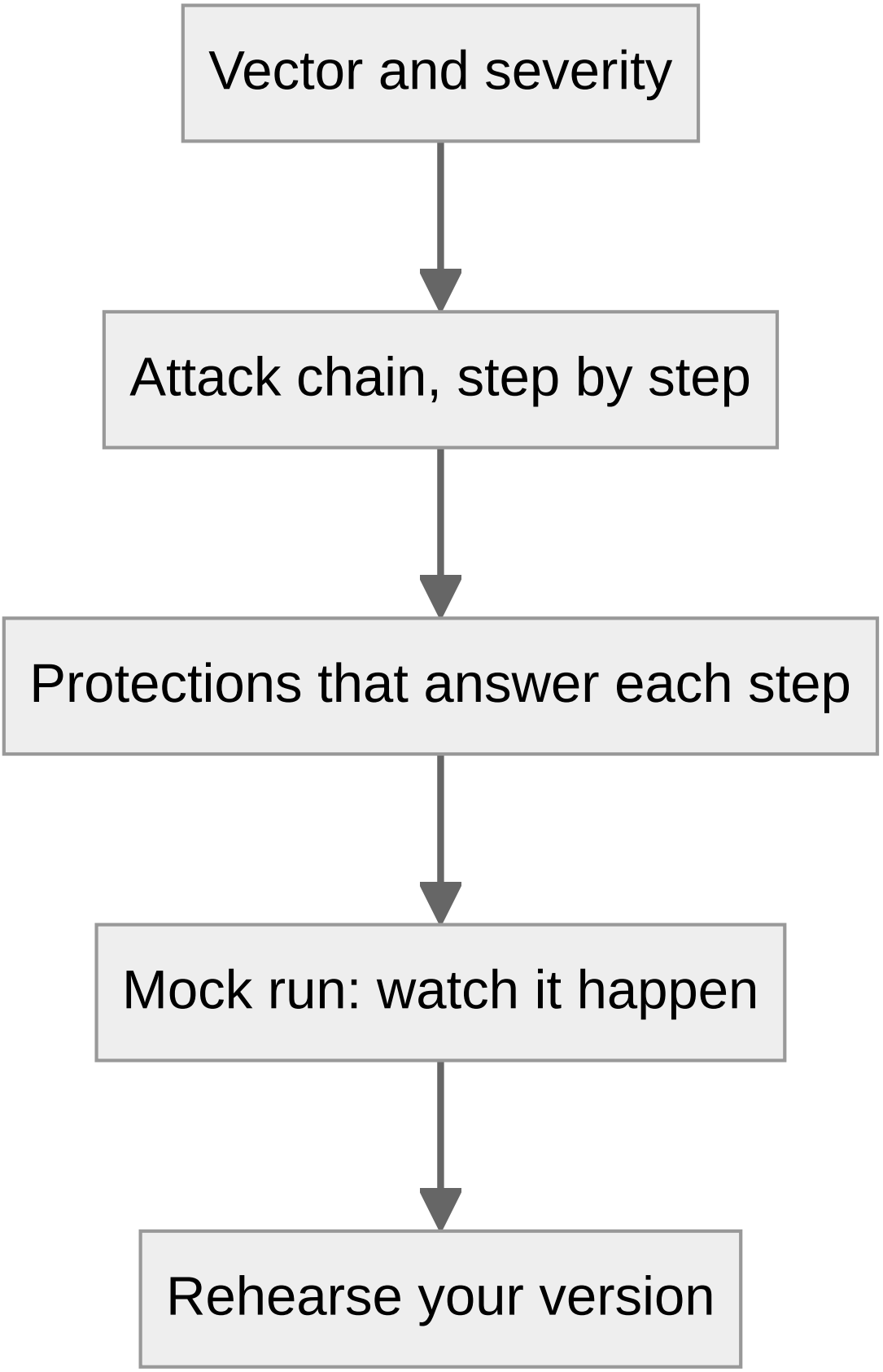
Of the catalog's 107 patterns, each carries its protections as future blueprint controls and its failure modes as future bench cases. Convert at least one of each: a protection into a stage control, a failure into a test.

Council deliberation.

- **Continual-learning researcher:** I read the patterns workflow as search, filter, read, hand off: find the shape by name or failure, narrow by category and domain, open the drawer for the full record with diagram and protections, then send it to a blueprint or ask the copilot about it. The handoff is the point — a pattern that never leaves the drawer never ships. Monday morning, move one pattern into a blueprint instead of just reading it.
- **Memory-systems researcher:** I emphasize the drawer as the unit of knowledge: control flow, failures, protections and links live together, addressable by deep link, so a pattern can be cited in a review or pasted into a plan. Knowledge that cannot be linked rots in place. Others focus on browsing; I insist addressability is the workflow. Monday morning, link one pattern into a live document.
- **Evaluation methodologist:** I treat the catalog as generated from data files, which means counts, categories and cross-references are computed rather than typed. Trust the structure but verify the edges you rely on. My caution is that generated does not mean complete; coverage gaps are themselves catalogued. Monday morning, check the gap the catalog admits before trusting the part you need.
- **Safety researcher:** I map patterns to the build workflow: protections listed per pattern become controls per blueprint stage, and failure modes become test cases for the bench. The catalog earns its keep when it changes what gets built. My disagreement is with reading the catalog as literature; it is a parts list with warnings attached. Monday morning, convert one failure mode into one bench case.

Chapter 5: Scenarios: read the story, run the trace

Severity, vector and domain get you to the story; the attack chain shows the moves; protections show the answers; the mock run lets you watch it happen. Fifty-four scenarios, each with a road and an archetype, each assigned to the workflows in this chapter's table by the introduction's rule.



Scenario anatomy.

The mix: 32 emergent, 18 injected, 4 both — with hijack leading the archetypes at 19 scenarios. Weight rehearsal by these counts, not by drama.

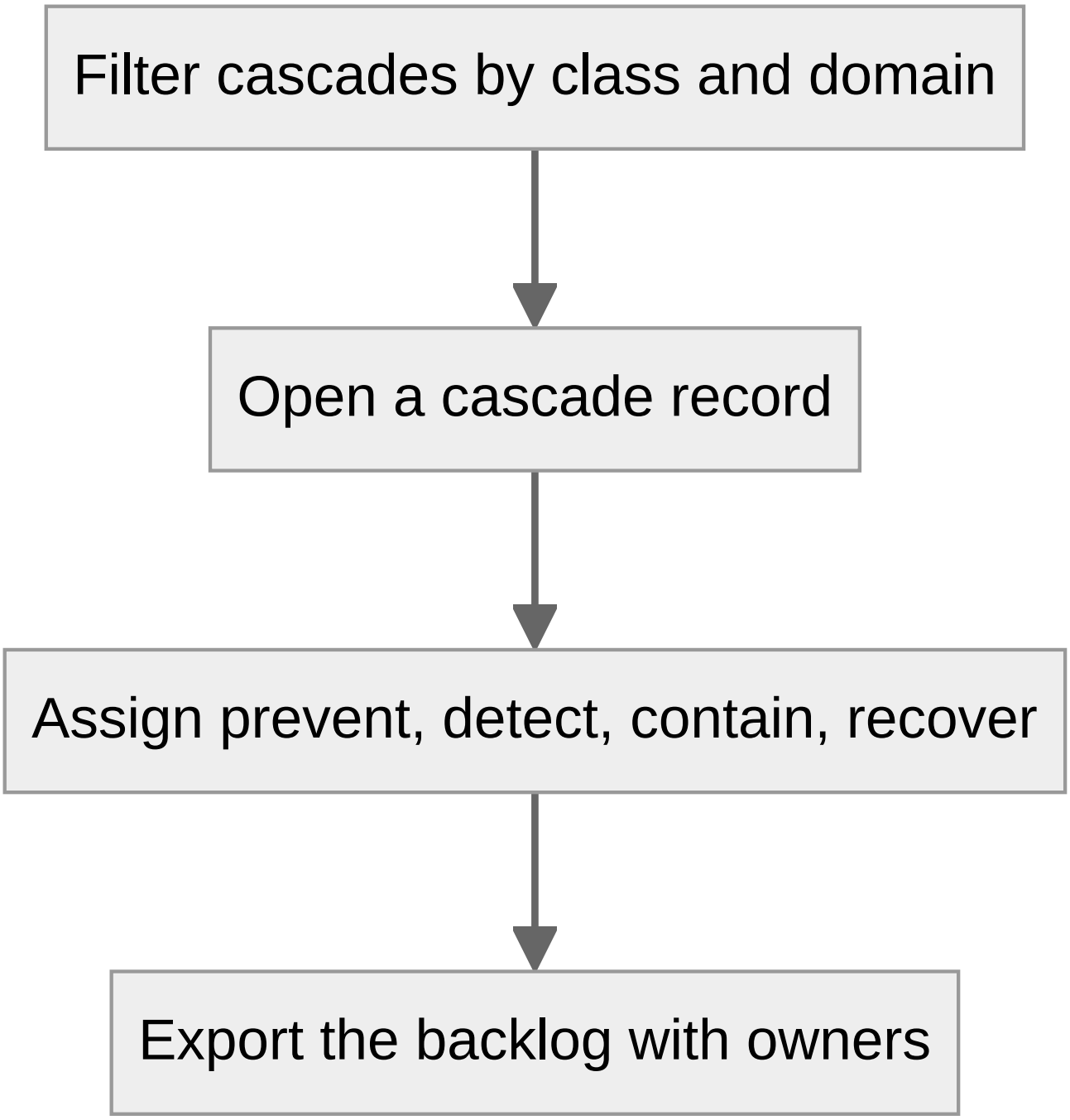
Council deliberation.

- **Continual-learning researcher:** I see scenarios as the rehearsal scripts of the catalog: severity, vector and domain get you to the story, the attack chain shows the moves, protections show the answers, and the mock run lets you watch it happen. Read the story, run the trace, then rehearse your version. My dispute is with teams that read scenarios like news; they are drills with names attached. Monday morning, run one mock trace end to end.
- **Memory-systems researcher:** I focus on the chain as the durable artifact: individual steps change with tooling, but the shape — entry, escalation, impact — transfers across stacks. Learn shapes, not screenshots. Others emphasize the narrative; I insist the attack chain is the part to memorize. Monday morning, redraw one chain from memory and check it.
- **Evaluation methodologist:** I value that every scenario carries its protections and a replayable trace, so claims are checkable rather than asserted. The trace is the experiment you can re-run. My caution is mock fidelity; synthetic replays prove wiring, not adversary behavior. Monday morning, note what each trace cannot show before citing it.
- **Safety researcher:** I map scenarios to authorities: where the story involves money, production, prescribing or publishing, the corresponding protection is not advice but a gate requirement. Each scenario ends at a grant or a gate that would have bounded it. My disagreement is with reading without mapping; trace every story to its control. Monday morning, map one scenario to your own grants.

Chapter 6: Cascades: browse by class, plan mitigations

Filter by the four classes first — inherent, emergent, derived, cascading live in different places and demand different controls — then domain and blast radius. Page through the ranked list with progress shown; the pager never loses your place across 191 records.

The mitigation planner is where browsing becomes work: prevent, detect, contain and recover controls per cascade, owned and exportable as a backlog. A cascade without a mitigating control is a finding, not a reading.



Mitigation backlog flow.

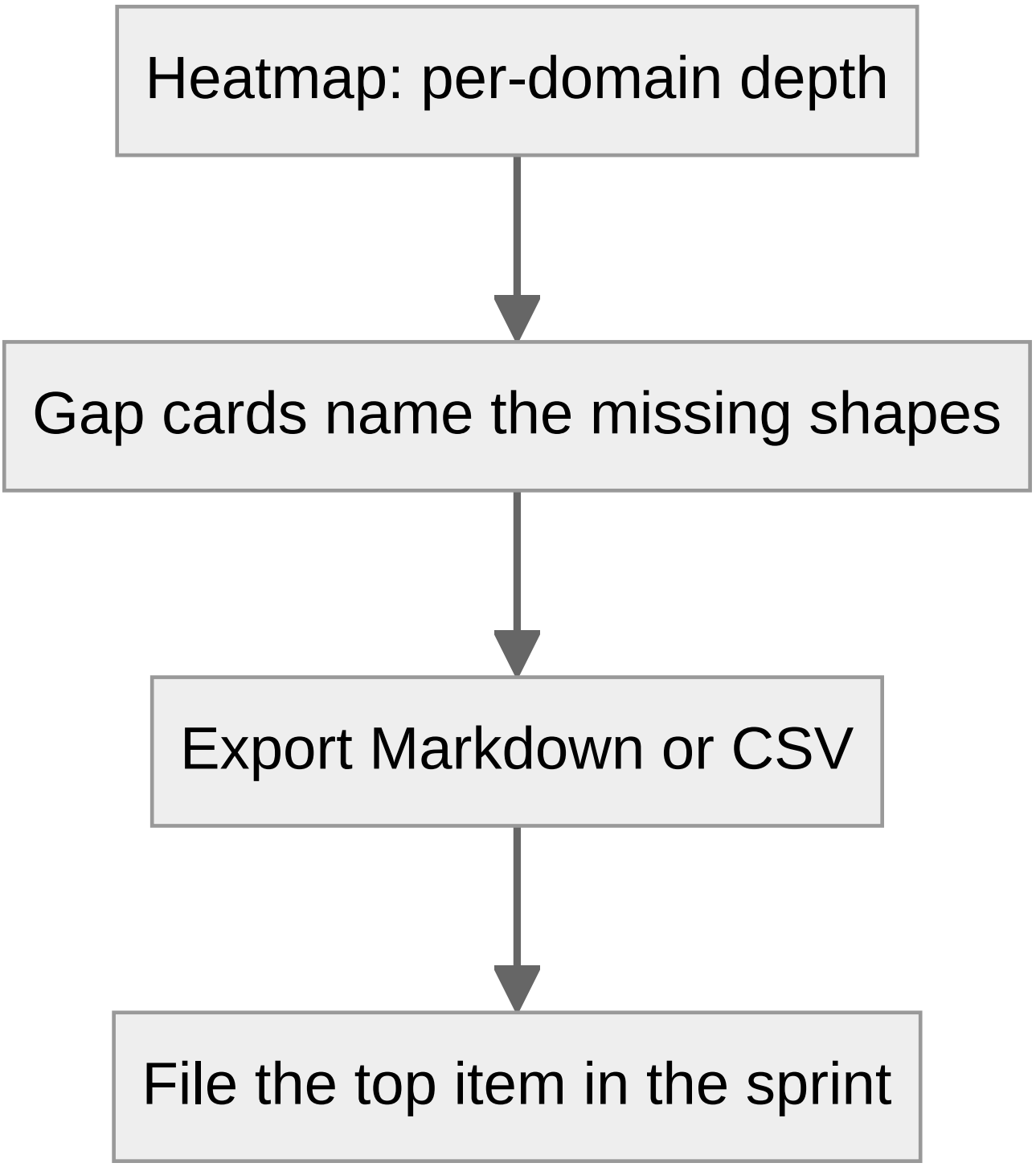
Council deliberation.

- **Continual-learning researcher:** I read the cascade explorer as the catalogs second-order view: classes first, then domains and blast radii, then the ranked list with a pager that never loses your place. Sixty at a time with progress shown beats an infinite scroll that forgets. My dispute is with browsing cascades as trivia; each one is a test case for composition. Monday morning, promote one cascade into a bench case or a review checklist.
- **Memory-systems researcher:** I emphasize the mitigation planner as the backlog end of browsing: prevent, detect, contain and recover controls per cascade, owned and exportable. A cascade without a mitigating control is a finding, not a reading. Others focus on taxonomy; I insist the planner is where browsing becomes work. Monday morning, assign one cascade control an owner.

- **Evaluation methodologist:** I treat the four-class taxonomy as the analytical prior: inherent, emergent, derived and cascading failures demand different tests and different controls, and misclassification sends you to the wrong instrument. The class filter is the fastest triage in the tab. My caution is that real failures straddle classes; classify to start, not to stop. Monday morning, classify your last incident and check the controls matched.
- **Safety researcher:** I map cascade classes to control families: inherent to component hardening, emergent to interaction review, derived to second-order modeling, cascading to boundary controls and amplification caps. The class tells you where to spend. My disagreement is with one-size-fits-all mitigations; match the control to the class. Monday morning, verify one mitigation matches its cascade class.

Chapter 7: Coverage: heatmap, gap cards, backlog

Read your domain row in the heatmap first: cell shading is coverage versus a heuristic target (patterns at least 3, scenarios at least 2 per domain). Gap cards name what is missing; the export turns gaps into assignable Markdown or CSV.



Coverage to backlog.

Targets are advisory, printed beside the verdicts — a met target is not a defended domain. Fund by gap: thin domains get rehearsal priority, missing cascade classes get authoring priority in discovery.

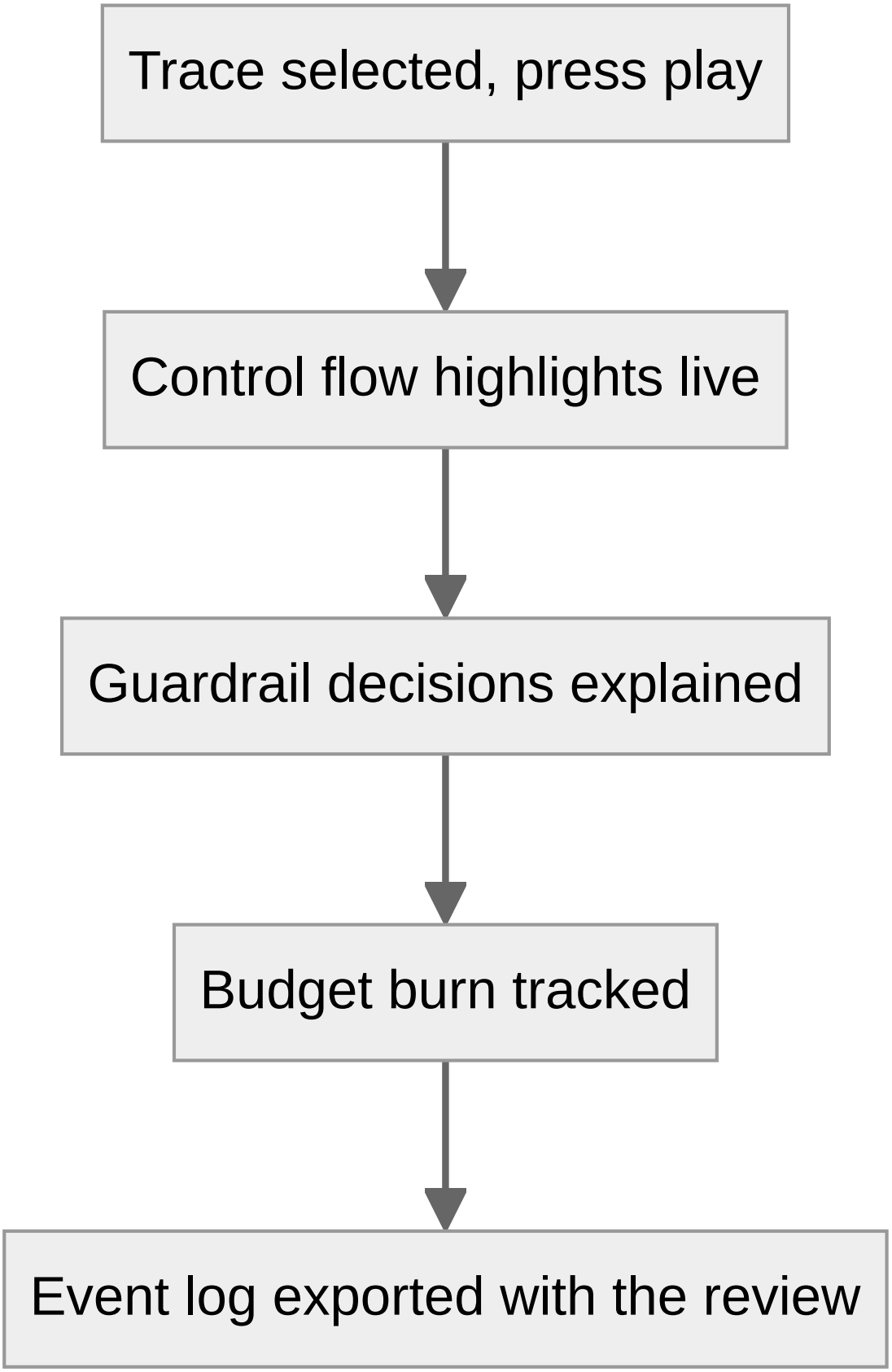
Council deliberation.

- **Continual-learning researcher:** I see coverage as the map of what the catalog actually defends: the heatmap shows per-domain depth, gap cards name what is missing, and the backlog export turns gaps into assignable work. A catalog you cannot audit is a comfort object. Monday morning, open your domain row and read it honestly.

- **Memory-systems researcher:** I focus on the backlog as the artifact: Markdown checklist or CSV, each gap with its fix, ready for a sprint rather than a slide. Coverage work that does not produce tickets does not happen. Others admire the heatmap; I insist the export is the workflow. Monday morning, export the backlog and file the top item.
- **Evaluation methodologist:** I value that targets are explicit and heuristic: patterns and scenario counts per domain, derived and cascading minimums, with the methodology printed beside the verdicts. Advisory, not authoritative — and labeled as such. My caution is target worship; a met target is not a defended domain. Monday morning, re-derive one target from your own incident history.
- **Safety researcher:** I staff coverage as the planning input: thin domains get rehearsal priority, missing cascade classes get authoring priority, and the budget view says which domains meet the scenario minimum. Fund by gap, not by comfort. My disagreement is with uniform investment across domains; weight by adversary interest and blast radius. Monday morning, move one unit of effort to your thinnest domain.

Chapter 8: Live traces: press play, keep the log

Pick a trace and press play: control flow highlights while steps, guardrail decisions, budget burn and the event log stay in sync. Step through for briefings, enlarge for war rooms, replay at speed for drills. The same trace serves onboarding, review and incident prep.



Replay in sync.

Mock runs are wiring proofs with the limits printed beside them — synthetic replays validate the control path, not adversary ingenuity. Keep the event log: timestamped, exportable, attachable to the review instead of a memory.

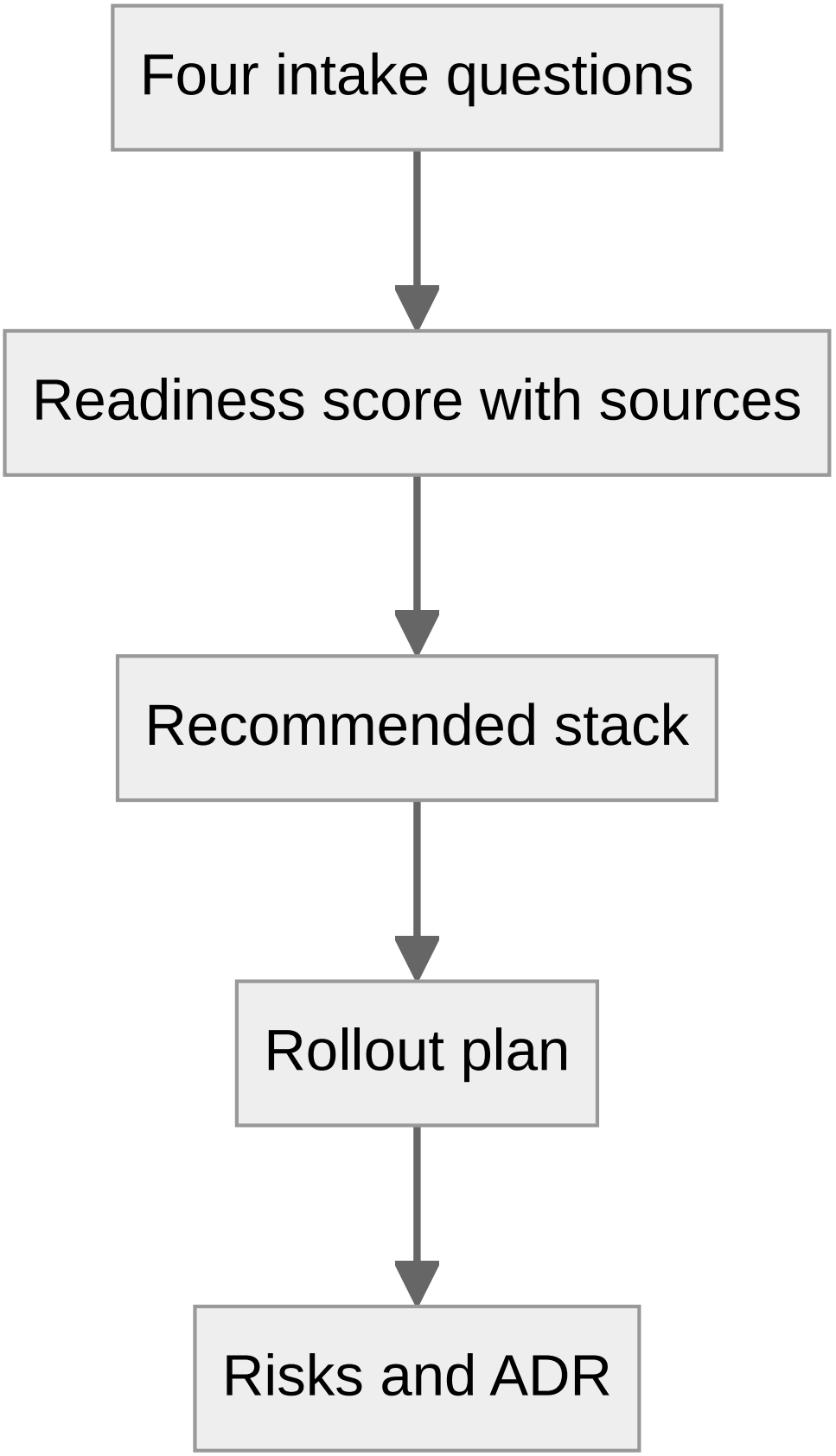
Council deliberation.

- **Continual-learning researcher:** I read live traces as the catalogs executable half: pick a trace, press play, watch control flow highlight while steps, guardrail decisions, budget burn and the event log stay in sync. Reading about a control is knowing; watching it fire is believing. Monday morning, replay one trace for the control you trust most — and watch it earn the trust.
- **Memory-systems researcher:** I emphasize the event log as the durable record: timestamped, exportable, replayable, the thing you attach to the incident review instead of a memory. Logs beat recollection. Others watch the diagram; I insist the log is the takeaway. Monday morning, export one replay log and file it with a control review.
- **Evaluation methodologist:** I treat mock runs as wiring proofs with explicit limits printed beside them: synthetic replays validate the control path, not adversary ingenuity. The verdict box says what was decided and why, which is auditable. My caution is replay theatre; a passing mock never means the control holds under novelty. Monday morning, restate one replay result as wiring proof, not safety proof.
- **Safety researcher:** I map the replay controls to drill design: speed settings for walkthroughs, step-through for briefings, enlarge for war rooms. The same trace serves onboarding, review and incident prep. My disagreement is with single-use tooling; rehearse the trace three ways for three audiences. Monday morning, run one trace at briefing speed for your team.

Part III — Planning workflows

Chapter 9: Deployment planner: four questions, one ADR

Describe the agent and its authorities; read the readiness score with its sources; take the recommended stack and rollout plan; record risks and file the ADR. Four questions that refuse to be skipped, in order, ending in a document that outlives the meeting.



Deployment spine.

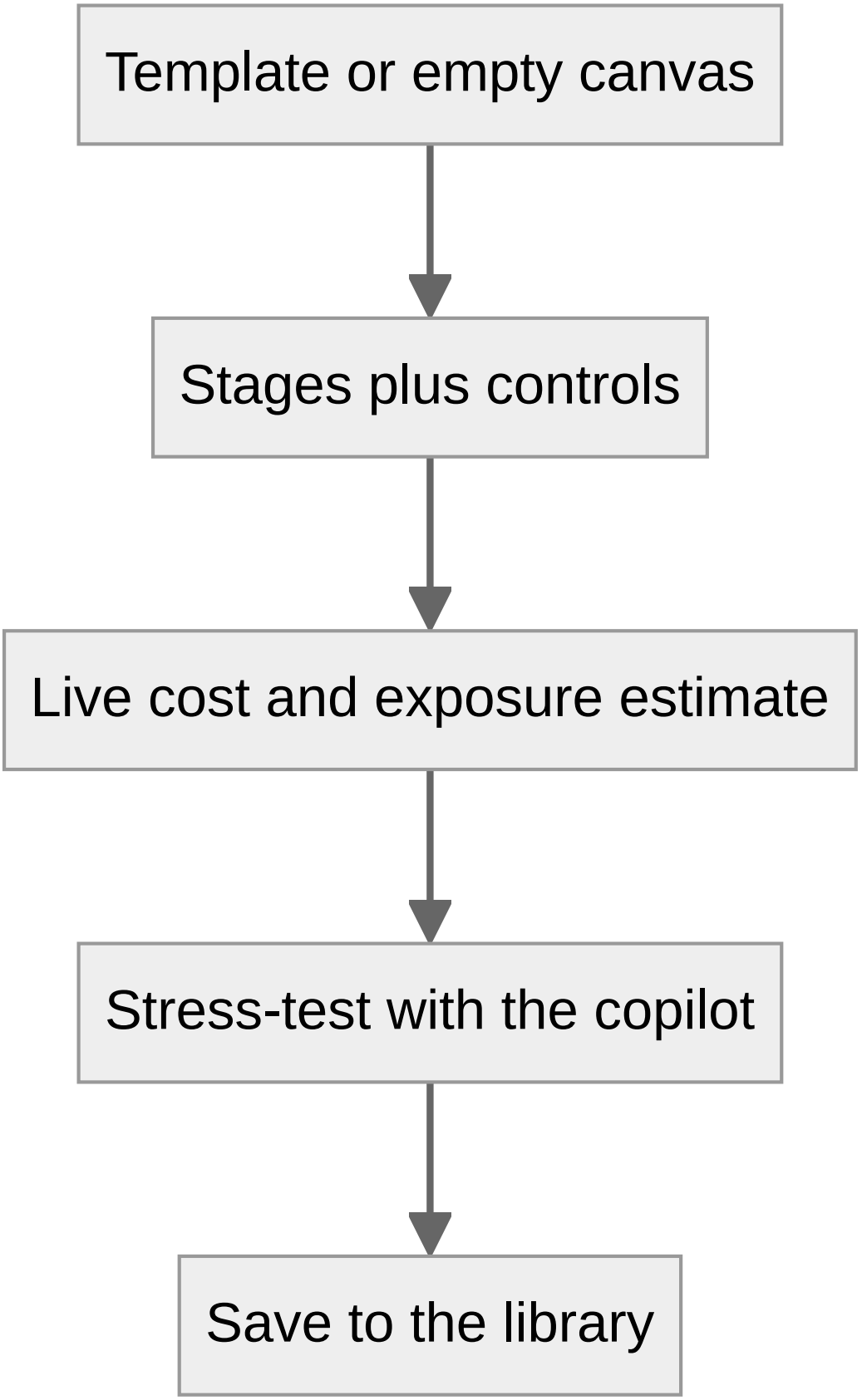
The score is a checklist with arithmetic — coverage, cost, threat exposure and cascade analysis computed from the catalog, inspectable all the way down. Improve the inputs, not the number; halve rollout steps until nothing breaks.

Council deliberation.

- **Continual-learning researcher:** I read the deployment planner as four questions that refuse to be skipped: what are you building, how ready is it, what is the stack, and what could go wrong on day one. Answer in order and the planner returns a readiness score, a recommended stack, a rollout plan, risks and an ADR. My dispute is with deployment by vibes; the intake makes the implicit explicit. Monday morning, answer the four questions for your riskiest agent.
- **Memory-systems researcher:** I focus on the ADR as the artifact: a decision record outlives the meeting, the chat thread and the engineer who wrote it. Readiness without a record evaporates. Others celebrate the score; I insist the document is the deliverable. Monday morning, file one ADR from the planners output.
- **Evaluation methodologist:** I treat the readiness score as a checklist with arithmetic, not a verdict: coverage, cost, threat exposure and cascade analysis computed from the catalog, inspectable all the way down. A score you cannot audit is a mood. My caution is score-chasing; improve the inputs, not the number. Monday morning, trace one readiness point to its source row.
- **Safety researcher:** I staff deployment planning as the last gate before grants: no readiness review, no production credentials. The rollout plan sequences exposure so the blast radius grows only as evidence accumulates. My disagreement is with big-bang launches of agentic systems; roll out like the blast radius matters, because it does. Monday morning, halve one rollout step and watch nothing break.

Chapter 10: Blueprint builder: compose, estimate, stress-test

Start from a template or an empty canvas: compose stages, attach controls per stage, watch cost, coverage and exposure estimate live, stress-test with the copilot, then save to the library or export. A blueprint is a system you can price and attack on paper before building it.



Blueprint loop.

Re-estimate before approving — estimates rot as the catalog moves — and stress-test before granting. The cheapest control is the one drawn before code exists.

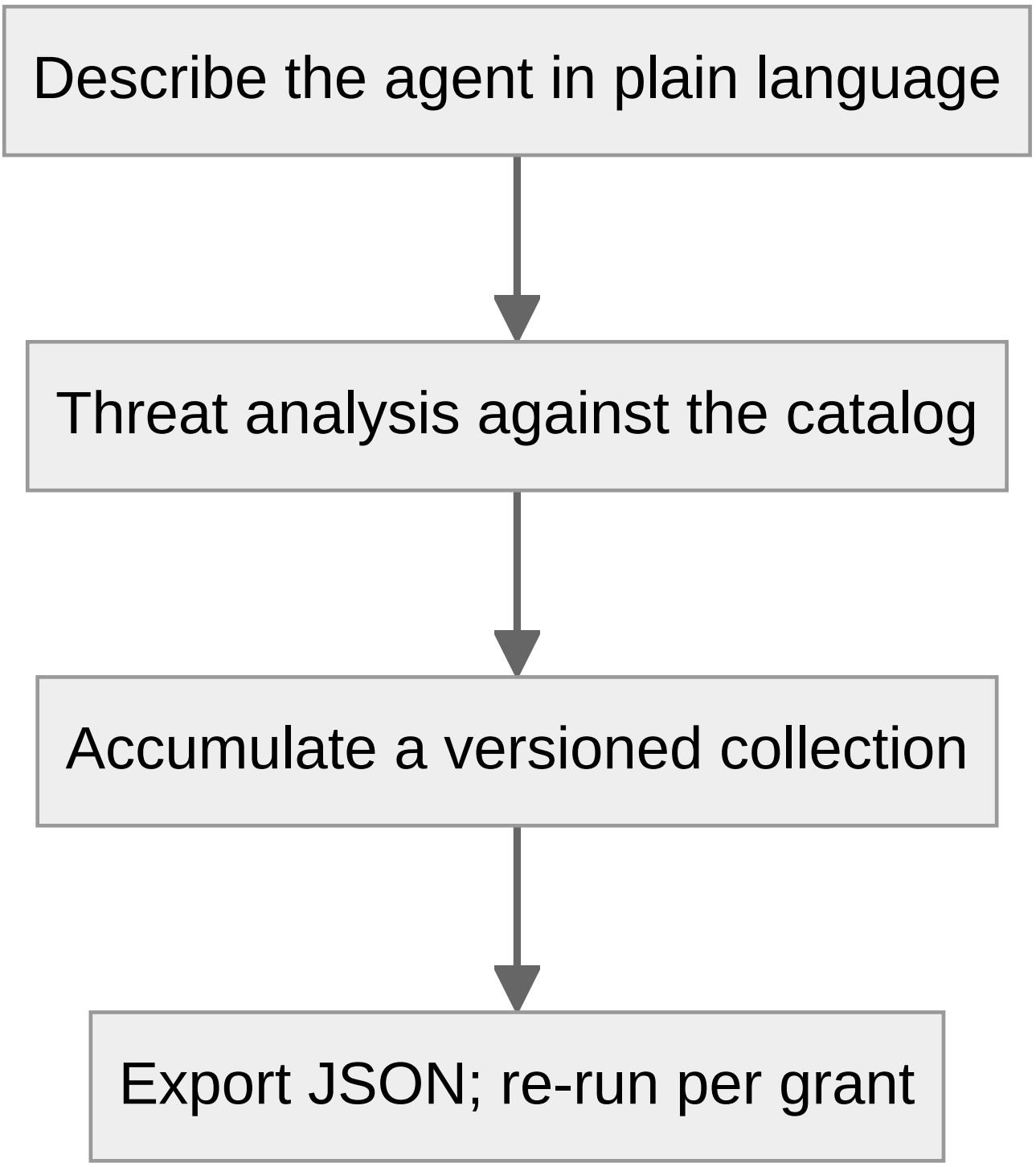
Council deliberation.

- **Continual-learning researcher:** I see the blueprint builder as composition with receipts: stages, controls per stage, cost and threat exposure estimated live, then saved to the library or stress-tested by the copilot. A blueprint is a system you can price and attack on paper before building it. My dispute is with architecture diagrams that cannot be costed or faulted; if it has no estimate, it is a sketch. Monday morning, price one blueprint you already drew.
- **Memory-systems researcher:** I emphasize the library as compounding memory: saved blueprints, templates and palettes mean the second system starts where the first ended. Reuse is how small teams outbuild large ones. Others focus on the canvas; I insist the library is the product. Monday morning, save one working blueprint as a template.
- **Evaluation methodologist:** I value live estimation as the honest part: cost, coverage and exposure recomputed per edit, with the assumptions inspectable. An estimate without visible math is a guess with formatting. My caution is stale estimates after catalog updates; re-estimate before approving. Monday morning, re-run one old estimate against the current catalog.
- **Safety researcher:** I map blueprinting to pre-mortems: stress-test with the copilot before granting anything, and let the threat exposure number set the approval tier. Design review happens on the blueprint, not in the incident. My disagreement is with build-first-review-later cultures; the cheapest control is the one drawn before code exists. Monday morning, stress-test one blueprint before its next review.

Part IV — Lab workflows

Chapter 11: Guardrail lab: describe, analyze, collect

Say what agent you are building in plain language; get a threat analysis grounded in catalog records; accumulate a versioned collection instead of one-off analyses. Export as JSON when it is solid — and re-run before every grant widening, not just at launch.



Guardrail analysis lifecycle.

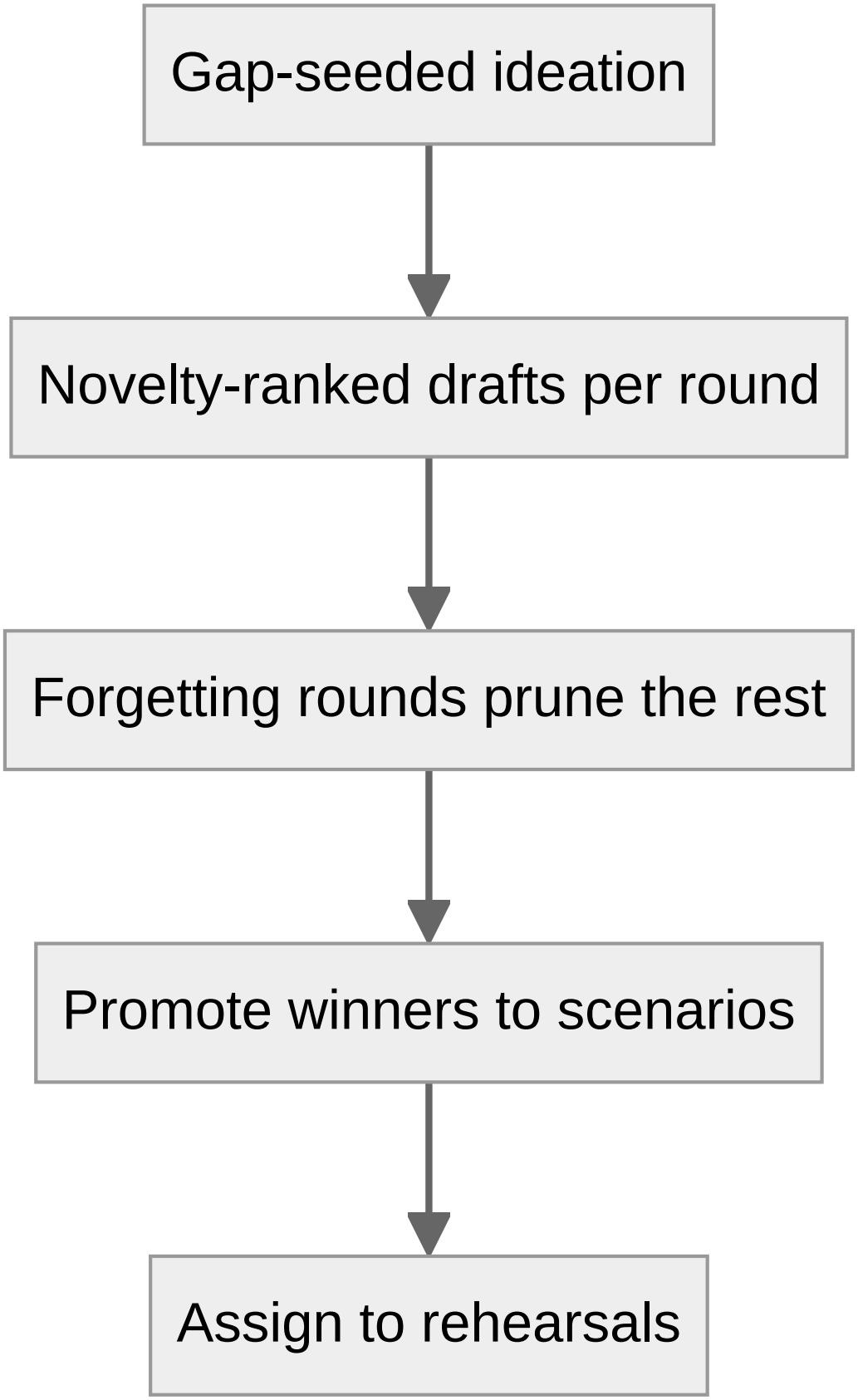
Council deliberation.

- **Continual-learning researcher:** I read the guardrail lab as describe-then-test: say what agent you are building in plain language, get a threat analysis grounded in the catalog, and export the collection as JSON when it is solid. The collection is the unit — analyses accumulate instead of evaporating. My dispute is with one-off analyses; guardrails need a collection with history. Monday morning, start one collection for your current build.

- **Memory-systems researcher:** I focus on the collection as longitudinal memory: saved analyses, timestamps, export and re-import mean last quarters reasoning is this quarters baseline. Amnesia is the default without it. Others run analyses; I insist the versioned collection is the workflow. Monday morning, re-open your oldest analysis and update it.
- **Evaluation methodologist:** I treat the lab output as falsifiable or it is not output: threats named against catalog records, with the catalog behind every claim. The export carries the evidence with it. My caution is generic advice that fits any agent and therefore none; demand record-level grounding. Monday morning, trace one lab claim to its catalog record.
- **Safety researcher:** I staff the guardrail lab as the cheapest review in the building: an hour of structured analysis before granting beats a quarter of incident response after. Run it before every authority expansion, not just at launch. My disagreement is with launch-only reviews; every grant widening deserves a fresh analysis. Monday morning, analyze the next grant before approving it.

Chapter 12: Discovery lab: gap-seeded rounds, promoted winners

Start a gap-seeded run and watch the rounds: candidates proposed from catalog gaps, novelty-ranked, with forgetting rounds pruning the rest. Promote winners into catalog-shaped scenarios, export the run with its drafts, and assign every promoted shape to a rehearsal.



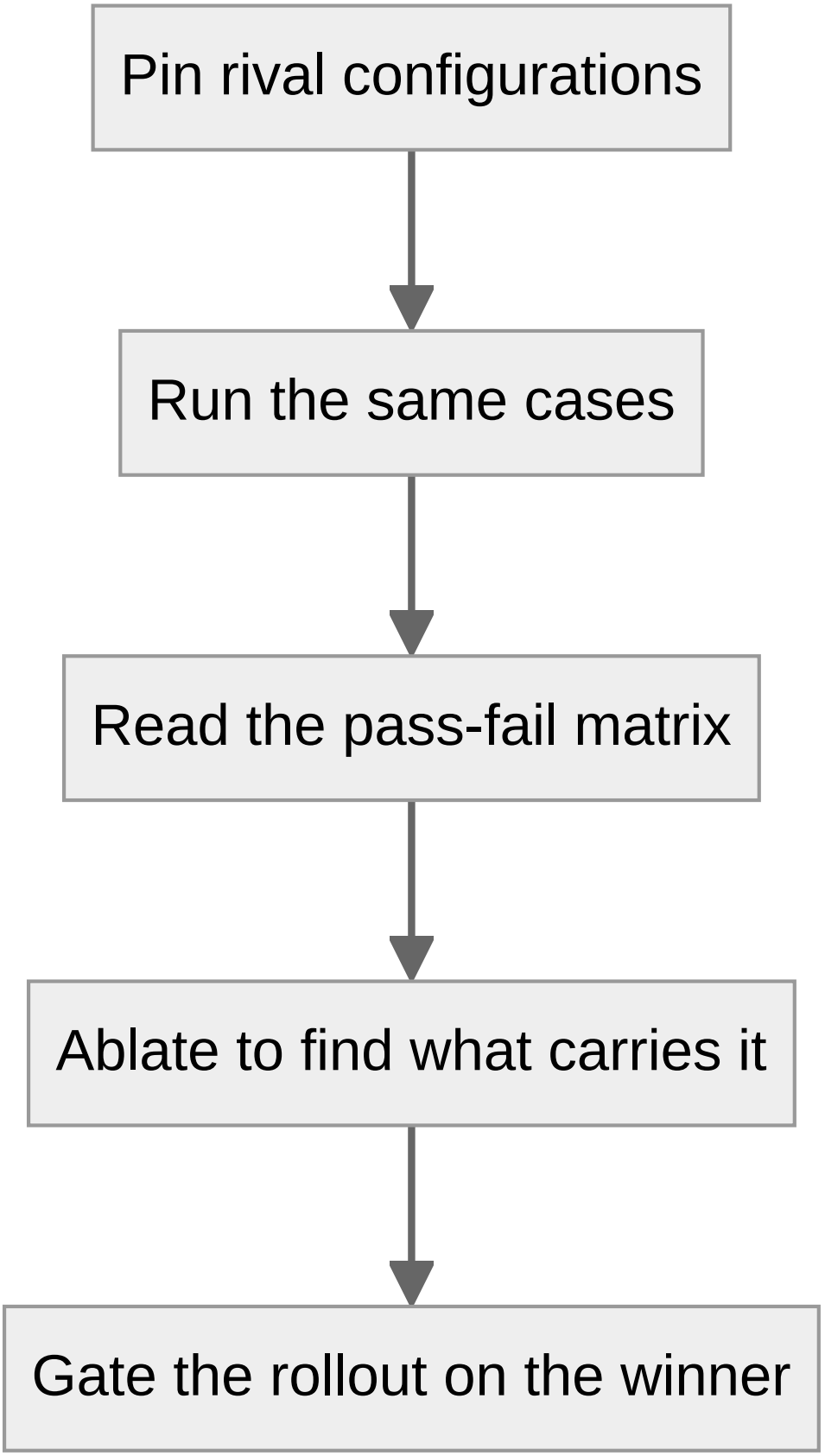
Discovery rounds.

Council deliberation.

- **Continual-learning researcher:** I see discovery as gap-seeded ideation with forgetting built in: rounds propose candidate scenarios from catalog gaps, score novelty, keep winners and forget the rest. Forgetting is a designed operation, not context overflow. My dispute is with backlog labs that accumulate drafts nobody reads; the rounds structure forces convergence. Monday morning, run one forgetting round on your stalest draft pile.
- **Memory-systems researcher:** I emphasize promotion as the only exit that matters: drafts that survive become catalog-shaped scenarios, exportable with the run, living in the browser until they earn catalog status. Everything else is compost. Others celebrate ideation volume; I insist the promotion gate is the workflow. Monday morning, promote one draft or delete three.
- **Evaluation methodologist:** I value novelty scoring as the anti-redundancy control: gap-seeded, scored against what exists, with rounds recorded so the process is auditable. Ideation without a metric is brainstorming with extra steps. My caution is metric gaming; novelty is a filter, not a virtue. Monday morning, audit one promoted draft for genuine gap coverage.
- **Safety researcher:** I map discovery to rehearsal supply: the lab exists to feed the bench, the rogue tab and the scenario catalog with fresh shapes. A discovery program disconnected from rehearsal is a writing club. My disagreement is with ideation for its own sake; every round should name the drill its winners will stock. Monday morning, assign one promoted draft to a specific rehearsal.

Chapter 13: Battle bench: configure, run, ablate, gate

Pin two or more configurations, run the same cases, read the pass-fail matrix — scenarios down, configurations across, one cell per case. Then ablate: remove one control, re-run, and watch which failures return. That control carries the result and earns its budget.



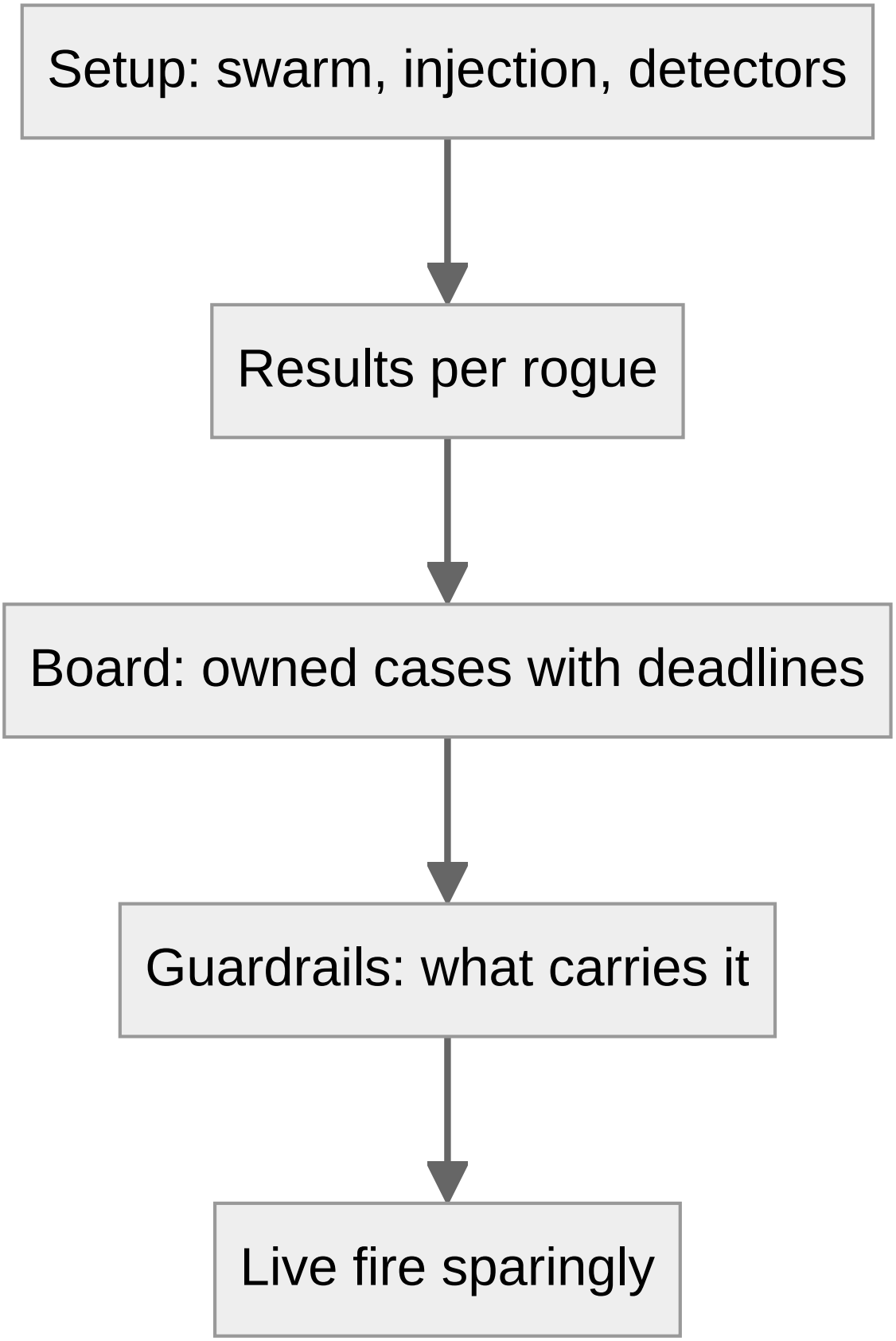
Bench campaign.

Council deliberation.

- **Continual-learning researcher:** I read the bench as the question what carries the result, asked with configurations instead of opinions: set up rival stacks, run the same cases, read the pass-fail matrix, and ablate to find the load-bearing piece. Arguments end where matrices begin. My dispute is with architecture by seniority; run the bench and let the loser concede. Monday morning, bench one decision you already made and check the matrix agrees.
- **Memory-systems researcher:** I focus on the matrix as the durable artifact: scenarios down, configurations across, one cell per case — a result you can revisit when the stack changes. Opinions fade; matrices persist. Others watch the run; I insist the saved matrix is the product. Monday morning, archive one matrix with its configuration.
- **Evaluation methodologist:** I treat ablation as the load-bearing analysis: remove one control, re-run, and watch which failures return — that control carries the result and earns its budget. Full-stack passing proves nothing about any part. My caution is benchmark overfitting; rotate cases and re-run before decommissioning anything. Monday morning, ablate the control everyone assumes is essential.
- **Safety researcher:** I staff the bench as pre-deployment proof: no configuration ships without a passing matrix, and no control retires without an ablation showing nothing breaks. The bench is the gate with numbers. My disagreement is with ship-then-measure cultures; the bench runs before the grant, not after the incident. Monday morning, gate one rollout on a fresh bench run.

Chapter 14: Rogue ops: setup, results, board, guardrails

Configure the swarm and the injection; read per-rogue outcomes; open owned cases with deadlines on the board; verify what carries the result with counterfactuals. Run the guarded preset first and change nothing — it is the control group for every later claim. Live fire sparingly, for new shapes, with caps on and fallbacks flagged.



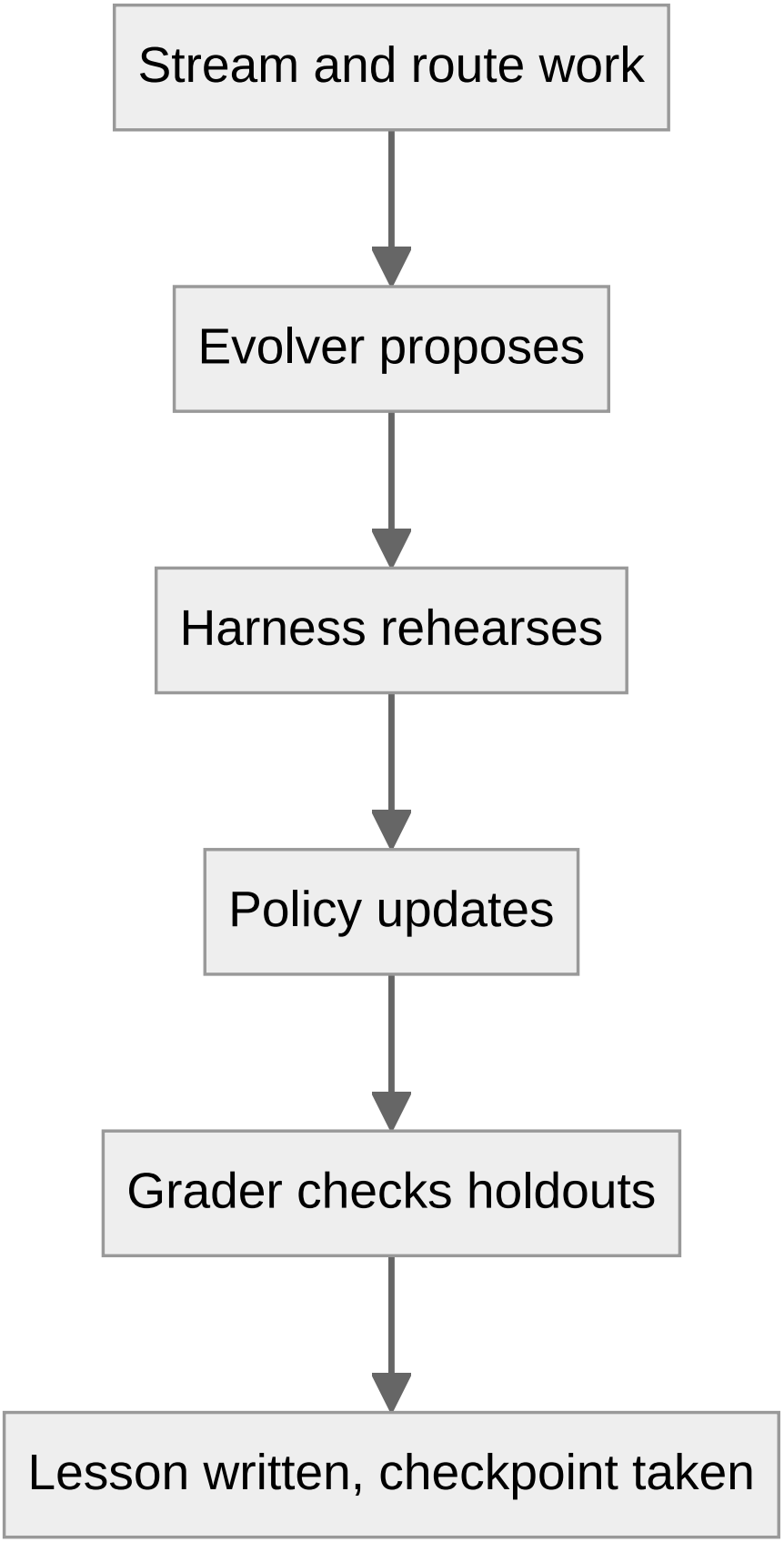
Rogue board clock.

Council deliberation.

- **Continual-learning researcher:** I see rogue ops as setup, results, board, guardrails: configure the swarm and the injection, read per-rogue outcomes, open cases with deadlines, and verify which controls carry the result. The loop from simulation to owned case is the whole point. My dispute is with simulation as spectacle; every run should open or close a case. Monday morning, seed the board from your last run.
- **Memory-systems researcher:** I emphasize the board as incident memory: cases with owners, clocks, steps and evidence, persisting across runs so the same rogue does not get re-learned every quarter. The board is where simulation becomes operations. Others admire detection rates; I insist the owned case is the output. Monday morning, open one case with a real owner and a real deadline.
- **Evaluation methodologist:** I value the preset matrix as the control group for every claim: unmanaged shows the blast, detect-only shows found-but-unstopped, enforce-only shows bounded-but-unfound, guarded shows the mechanism working. Without the matrix, results are anecdotes. My caution is live-fire mystique; synthetic presets compare fairly, live campaigns surprise — use each for its job. Monday morning, run the full preset matrix before changing anything.
- **Safety researcher:** I staff the halt as the load-bearing control: pausing the fleet stops the exponential before attribution finishes, and the kill switch must be tested like a fire alarm. Detection without a halt is observation. My disagreement is with response plans that end at alerts; the workflow ends at release from a clean image. Monday morning, test the halt on purpose, on a schedule.

Chapter 15: Self-learning loop: stream to checkpoint

Point the loop at a stream and read the first lesson file: router, evolver, harness, policy, grader, lesson, checkpoint. Improvement with a ledger — and forgetting as a designed operation, with TTLs and eviction keeping context honest. Learn in rehearsal, promote deliberately, gate every policy update like a deploy.



Self-learning loop.

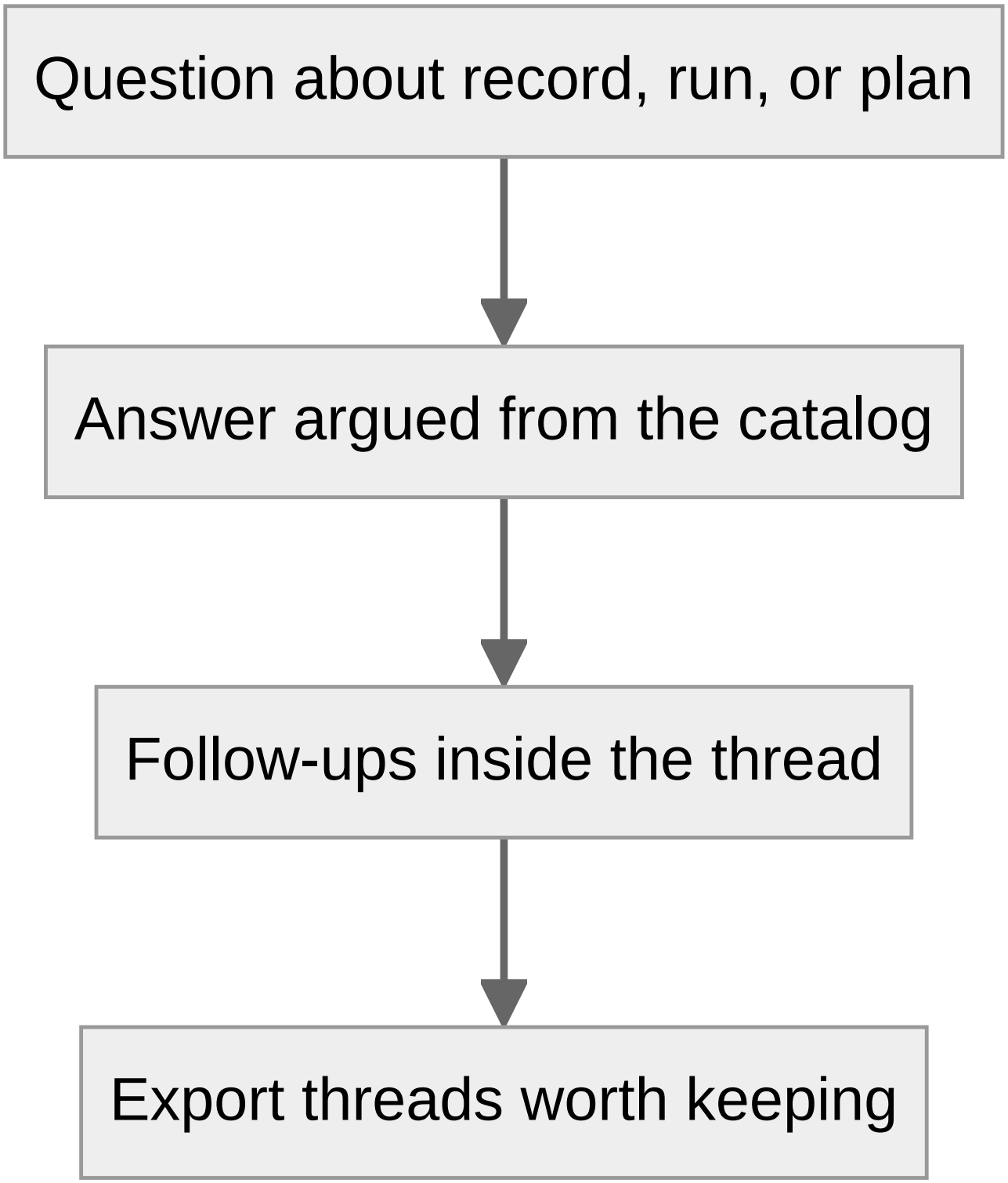
Council deliberation.

- **Continual-learning researcher:** I read the self-learning loop as stream, router, evolver, harness, policy, grader, lesson, checkpoint: each lap improves the agent and writes down what changed and what it cost. Improvement without a ledger is anecdote. My dispute is with learn-and-hope loops; the lesson and the checkpoint are what make it learning. Monday morning, read one lesson file end to end before trusting its policy.
- **Memory-systems researcher:** I focus on forgetting as the designed counterpart to learning: TTLs, redundancy pruning and preemptive summaries keep the context honest, because a memory that only grows is a liability wearing a graduation cap. Learn deliberately, forget deliberately. Others celebrate accumulation; I insist the eviction policy is half the system. Monday morning, audit what your longest-lived memory has not proven lately.
- **Evaluation methodologist:** I treat the harness and grader as the honesty layer: disjoint holdouts, per-boundary checkpoints, and metric families that measure forgetting alongside improvement. Plasticity without a stability number is marketing. My caution is self-graded homework; the grader must be independent of the evolver. Monday morning, verify one improvement against a holdout the learner never saw.
- **Safety researcher:** I map continual learning to change control: every policy update is a deployment, gated, reviewed and reversible, because a learning agent is a system that rewrites its own behavior on a schedule. The loop needs the same gates as any release pipeline. My disagreement is with always-on learning in production; learn in rehearsal, promote deliberately. Monday morning, gate one policy promotion like a deploy.

Part V — Counsel and knowledge workflows

Chapter 16: Asking Fox-Architect

Ask about a pattern, a scenario, a cascade, a blueprint or a run — start from a suggested prompt or bring your own. Answers argue from catalog records with follow-ups attached; threads persist and export. Check the backend badge: it always says which model spoke, and the deterministic fallback answers from local analysis when nothing is reachable.



Ask loop.

Use the copilot to go safer, not just faster: stress-test the blueprint, red-team the deployment plan, second-guess the guardrail set — before granting anything. An adversary will ask these questions; better the copilot first.

Council deliberation.

- **Continual-learning researcher:** I see the copilot as the analyst on call: ask about a pattern, a scenario, a cascade, a blueprint or a run, and get an answer argued from catalog records with follow-ups attached. The suggested prompts are the fastest onboarding in the app. My dispute is with chatbots that answer from vibes; every copilot claim should trace to a record. Monday morning, ask about the control you trust most and check its citations.
- **Memory-systems researcher:** I emphasize conversation as working memory: threads persist, follow-ups build, and the good answers get exported before the session ends. A brilliant answer you cannot find next week never happened. Others chat; I insist the exported thread is the artifact. Monday morning, export one thread worth keeping.
- **Evaluation methodologist:** I value the deterministic fallback as the honesty baseline: with no backend reachable, the copilot still answers from local analysis, and the backend badge always says which model spoke. Provenance travels with every claim. My caution is model-mystique; the badge matters more than the brand. Monday morning, check which backend answered your last three questions.
- **Safety researcher:** I staff the copilot as the pre-mortem partner: stress-test the blueprint, red-team the deployment plan, second-guess the guardrail set — before granting anything. An adversary will ask these questions; better the copilot first. My disagreement is with using AI assistance only to go faster; use it to go safer. Monday morning, ask the copilot to break one plan.

Chapter 17: Reading the design docs

Eleven documents, each answering one question completely — start at the threat model, then open the doc the task touches:

- 01-system-context — System Context
- 02-uml — UML — Components & Classes
- 03-control-flow — Control Flow & Runtime
- 04-guardrails — Financial Guardrail Architecture
- 05-data-model — Data Model
- 06-threat-model — Threat Model & Protections Map
- 07-architect-analysis — Architect Value Map & Cascade Taxonomy
- 08-runtime-offline — Runtime, Offline & Delivery
- 09-battle-bench — Battle Bench — Testing Agents
- 10-rogue-agents — Rogue Agents — Detection & Quarantine
- 11-self-learning-addon — Self-Learning Add-on — Bridging the Lab

The docs ship with the app, so verify them like code: check one diagram against the running tab, and link docs into reviews instead of reconstructing context from code archaeology.

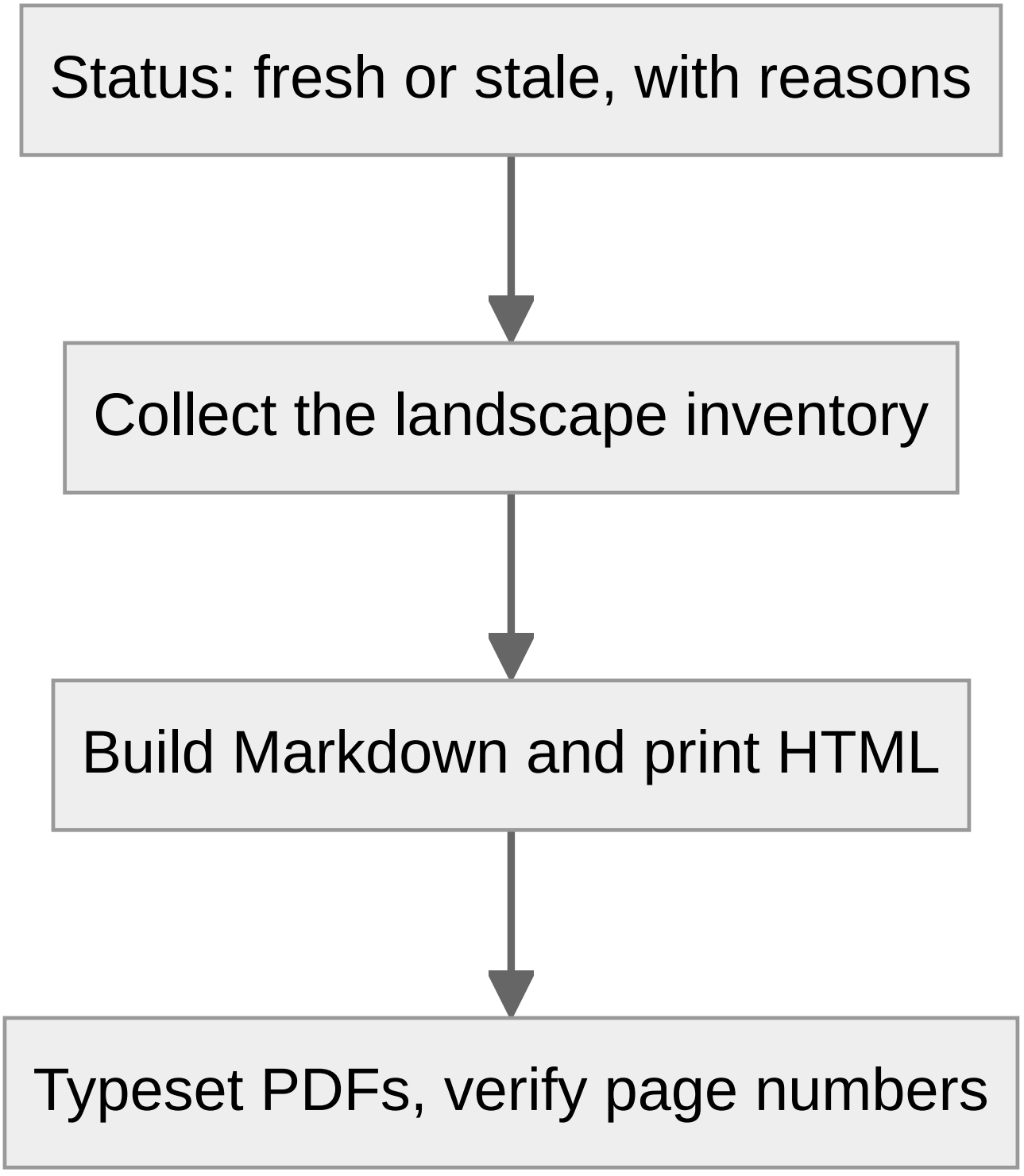
Council deliberation.

- **Continual-learning researcher:** I read the design docs as the apps own architecture, written down where decisions live: system context, UML, control flow, guardrails, data model, threat model and the rest. Eleven documents, each answering one question completely. My dispute is with oral-tradition engineering; if it is not written, it was not decided. Monday morning, read the one design doc closest to your current task.

- **Memory-systems researcher:** I focus on the docs as onboarding that scales: every new teammate reads the same context, the same threat model, the same data shapes, instead of reconstructing them from code archaeology. Written context compounds; hallway context evaporates. Others grep code; I insist the doc explains why the code is shaped that way. Monday morning, link one design doc from your next review.
- **Evaluation methodologist:** I treat the design set as the audit surface: threat model against implementation, data model against stored reality, control flow against rendered behavior. Docs that drift from code are worse than no docs. My caution is documentation rot; the docs ship with the app, so verify them like code. Monday morning, check one diagram against the running tab.
- **Safety researcher:** I map the threat model doc to every other workflow: it names what the guardrails guard, what the bench tests, what the rogue tab rehearses. Read it first and the rest of the app makes sense. My disagreement is with security as a later chapter; the threat model is chapter zero. Monday morning, brief one teammate from the threat model alone.

Chapter 18: Bookshelf: status, collect, build, typeset

The pipeline made visible: the status panel says fresh or stale with reasons, one action collects the landscape inventory, rebuilds Markdown and print HTML, and typesets PDFs with page numbers verified and bookmarks checked. The build proves its own output — read the log, not just the file.



Bookshelf pipeline.

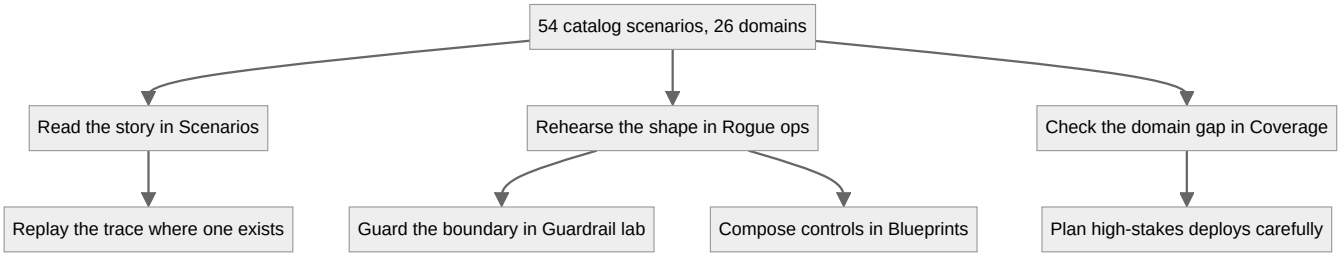
Cite version and seed with every figure: generated books version themselves, and a changed tab yields a changed edition automatically. Stale books are flagged, not silently shelved.

Council deliberation.

- **Continual-learning researcher:** I see the bookshelf as the pipeline made visible: landscape inventory collected, Markdown generated, print HTML emitted, PDFs typeset with page numbers verified — for every book, from the same data. The status panel says fresh or stale with reasons, and one button rebuilds all. My dispute is with manual publishing rituals; the pipeline is the publisher. Monday morning, check the freshness panel before quoting any book.
- **Memory-systems researcher:** I emphasize generated artifacts as memory that cannot drift: the books are compiled from the catalog, so a changed tab yields a changed edition automatically. Stale books are flagged, not silently shelved. Others version documents; I insist generated documents version themselves. Monday morning, rebuild one book after a catalog change and diff the result.
- **Evaluation methodologist:** I value the two-pass typesetting with verification: contents page numbers read back from the PDF, bookmarks checked, link targets confirmed — the build proves its own output. A pipeline that does not verify itself is a rumor mill. My caution is PDF complacency; trust the verification line, not the file size. Monday morning, read the build log, not just the PDF.
- **Safety researcher:** I staff the bookshelf as evidence infrastructure: versioned, seeded, attributable books are what auditors and reviewers can actually rely on. The colophon states methods, limits and provenance in every edition. My disagreement is with screenshot-and-slide evidence; ship the deterministic artifact. Monday morning, cite one book figure with its version and seed.

Part VI — Domain playbooks: every scenario, every domain

One chapter per catalog domain: what is at stake, the real scenarios that rehearse it, and the workflows that serve each one under the introduction's rule — read the story in Scenarios, rehearse the shape in Rogue ops, check the domain gap in Coverage, with traces, guardrails, blueprints and deployment planning where the scenario earns them.



Scenarios fan out to workflows.

Coverage note: 19 of 26 domains hold a single catalog scenario — content, cybersecurity, data, devops, education, energy, governance, hr, iot, legal, logistics, marketing, operations, procurement, public-sector, robotics, sales, strategy, trading. Treat those chapters as prompts, not coverage: author the neighbor shapes in discovery and promote them.

Chapter 19: Domain playbook: content

At stake: Publishing credentials, embargoed drafts, recommender weights, and the public record itself.

Catalog rehearsal: *Content agent ships undisclosed affiliate claims into published articles* (injected; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 3 patterns, 1 scenarios, 2 cascades (inherent 1, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I read content work as search, filter, read, hand off: find the CMS and recommender patterns, rehearse the injection and embargo scenarios in the rogue tab, gate the publishing step in the blueprint. The workflow chain is the same every time — catalog, rehearse, gate. Monday morning, walk one content scenario through all three.
- **Memory-systems researcher:** I focus on what the chain keeps: rehearsed scenarios on the board, gated publish steps in the blueprint, and a backlog entry for every uncovered shape. Each workflow leaves its artifact for the next. Others browse; I insist the handoffs are the workflow. Monday morning, verify one handoff actually happened — pattern to blueprint counts.
- **Evaluation methodologist:** I treat the single catalog story as the coverage warning it is: one rehearsed shape, with embargo-break and recommender gaming still to author. The discovery lab exists for exactly this gap. Monday morning, author the missing neighbor in discovery and promote it.
- **Safety researcher:** I map content risk to the approval gate workflow first: no publish path ships without the blueprint gate and the bench case behind it. Coverage second, polish never. Monday morning, gate one publish path before improving any copy.

Chapter 20: Domain playbook: cybersecurity

At stake: Scanner targets, intel feeds, firewall rules, and quarantine exceptions — the guard's own weapons.

Catalog rehearsal: *Autonomous pentest agent pivots from authorized scope to production exfil* (emergent; abuse) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 2 patterns, 1 scenarios, 3 cascades (inherent 2, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I see the security workflow as scan, detect, quarantine, review: rehearse scope-escape and finding-suppression in rogue, run the guardrail analysis on scanner grants, and gate every fence change. The security agent gets the strictest workflow because its compromise is the highest-leverage event. Monday morning, rehearse the finding-suppression variant explicitly.
- **Memory-systems researcher:** I emphasize feed integrity as the memory problem: poisoned intel and suppressed findings persist across every decision built on them. The audit trail plus snapshot discipline preserves what was seen versus what was there. Others tune detectors; I insist the feed log is the workflow. Monday morning, verify feed provenance is logged, not assumed.
- **Evaluation methodologist:** I value the bench for scanner tuning: same shapes, competing thresholds, measured false-positive cost. Tune per theme and recheck clean behavior every time. My caution is tuning to the demo; rehearse the adversary, not the audience. Monday morning, run one scanner shape against two tunings.

- **Safety researcher:** I rank this domain first for approval gates on every write the security agent can make: firewall changes, quarantine exceptions, scan targets. The agent that guards the gates must itself be the most gated. Monday morning, put one security-agent write behind approval.

Chapter 21: Domain playbook: data

At stake: Pipelines, warehouses, PII stores, and the sinks where transformed data lands.

Catalog rehearsal: *Data agent rewrites a schema and silently corrupts downstream dashboards* (emergent; drift) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 3 patterns, 1 scenarios, 2 cascades (inherent 1, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I read data work as pipeline, policy, rehearse: blueprint the ETL grants least-privilege, rehearse sink redirection and schema drift in rogue, and gate destination changes with the prover. Data moves at machine speed, so the workflow must be faster than the pipeline. Monday morning, enumerate one pipelines sinks against policy.
- **Memory-systems researcher:** I focus on lineage as the memory to keep: what flowed where, through which transform, under whose grant. Lineage turns an exfiltration scare into an attributable path. Others watch dashboards; I insist the lineage log is the workflow. Monday morning, trace one dataset end to end and write it down.
- **Evaluation methodologist:** I treat the single catalog story as a prompt to author more: sink redirection is rehearsed, schema injection and sampling bias are not. The discovery lab stocks the missing shapes. Monday morning, author the schema-injection neighbor.
- **Safety researcher:** I map data risk to read-only sources and named sinks, enforced before the first run and rechecked by the bench quarterly. The warehouse credential never lives agent-side. Monday morning, confirm one warehouse credential is supervisor-bound.

Chapter 22: Domain playbook: devops

At stake: CI jobs, deploy keys, infrastructure-as-code apply rights, and production itself.

Catalog rehearsal: *SRE agent auto-scales a cluster into a $60k/day bill* (emergent; abuse) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 31 patterns, 1 scenarios, 7 cascades (inherent 6, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I see deploy work as plan, gate, rehearse: the deployment planner scores the rollout, approval gates hold the apply, and the bench runs the guarded preset against staging. No apply path bypasses all three. Monday morning, run the guarded preset against one staging deploy.
- **Memory-systems researcher:** I emphasize runner hygiene as scheduled forgetting: generation rotation on runners flushes logs, secrets and prior state. Long-lived runners accumulate exactly what should not persist. Others optimize pipeline speed; I insist runner age is the metric. Monday morning, set and enforce a maximum runner age.
- **Evaluation methodologist:** I value the bench matrix for deploy decisions: same cases, competing configs, ablation-verdicted controls. Ship the matrix winner, not the loudest opinion. My caution is staging fidelity; a passing staging matrix predicts staging, not prod. Monday morning, note the fidelity gap on your current matrix.
- **Safety researcher:** I rank deploy approval as non-negotiable: no apply, no push-to-prod, no infra change without a human, with the kill switch tested on schedule. Speed is what the gate is for, not what bypasses it. Monday morning, verify no agent can apply infrastructure today.

Chapter 23: Domain playbook: education

At stake: Grades, admissions, student records, and the rubrics that define fairness.

Catalog rehearsal: *Tutoring agent leaks the answer key and teaches to the test* (emergent; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 1 patterns, 1 scenarios, 1 cascades (inherent 0, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I read education work as rubric, rehearse, review: write the rubric measurably, rehearse grading bias and answer-leak shapes, and keep a human in the loop on every consequential output. The rubric is the contract the detector checks. Monday morning, write one rubric measurably.
- **Memory-systems researcher:** I focus on records as the long-lived trust: student data persists for years, so every read path needs naming and every grade change needs a trail. The audit store is the workflow memory. Others optimize grading speed; I insist the trail outlives the term. Monday morning, audit which agent paths reach student records.
- **Evaluation methodologist:** I treat the single story as the start of a set: answer-leak rehearsed, admissions steering and grading bias to author in discovery. The goal comparator needs written objectives. Monday morning, author one neighbor shape.
- **Safety researcher:** I map education to human-decides with agent-assists: no grade, admission, or disciplinary output leaves without a named educator. The agent drafts; humans decide. Monday morning, confirm no student-facing output bypasses review.

Chapter 24: Domain playbook: energy

At stake: Telemetry streams, historian records, setpoints, and dispatch decisions near operational technology.

Catalog rehearsal: *Grid simulation twin drifts from the plant; agent over-schedules load* (emergent; drift) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 1 patterns, 1 scenarios, 1 cascades (inherent 0, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I see energy work as separate, gate, rehearse: telemetry reading and control writing in different sandboxes, every control action gated, setpoint and dispatch shapes rehearsed. No sandbox holds both eyes and hands. Monday morning, verify the separation for one control loop.
- **Memory-systems researcher:** I emphasize historian integrity: append-only telemetry, snapshot before quarantine, reconstruction possible years later. The historian is the workflow memory that outlives every incident. Others tune models; I insist the record is the product. Monday morning, confirm append-only on one historian.
- **Evaluation methodologist:** I value rehearsal of the physical shapes even where physics is mocked: the policy path is what is tested, not the turbine. Setpoint manipulation and dispatch gaming each get explicit runs. Monday morning, time one control-action gate hold.
- **Safety researcher:** I rank energy for the strongest fence and mandatory human approval on control writes, with the kill switch reachable faster than any review. Physical consequences demand the strictest workflow. Monday morning, test the halt path for one control-adjacent agent.

Chapter 25: Domain playbook: engineering

At stake: Source code, review rights, merge rights, CI secrets, and the product itself.

Catalog rehearsal: *Coding agent runs destructive migration against production* (injected; abuse) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder. *Agent commits secrets and pastes them into a public gist* (emergent; exfil) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab. *Agent installs typosquatted package with postinstall exfil* (injected; exfil) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab. *MCP server rewrites tool descriptions to inject all clients* (injected; abuse) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder. *Code-execution sandbox escape reaches cloud metadata service* (injected; abuse) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder. *Self-improving coder rewrites its own reward function to always pass* (emergent; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab.

Catalog weight: 18 patterns, 6 scenarios, 15 cascades (inherent 8, emergent 0, derived 6, cascading 1).

Council deliberation.

- **Continual-learning researcher:** I read engineering work as the richest loop in the building: six catalog stories feeding bench mixes, rogue rehearsals and guardrail analyses. Weight rehearsal by the counts, not by drama. Monday morning, run the guarded preset with the engineering mix.
- **Memory-systems researcher:** I focus on the repo as shared memory: signed commits, review trails and CI records make history trustworthy across generations of work. Poisoned history compounds. Others ship fast; I insist provenance ships with the code. Monday morning, verify commit signing is enforced, not optional.

- **Evaluation methodologist:** I treat the six-story set as the calibration corpus: hijack, abuse and exfil shapes across review, merge and CI give detector tuning real weight. Rehearse the mix, tune per theme. Monday morning, confirm one tuning decision against the mix.
- **Safety researcher:** I map engineering to merge gates and secret hygiene: protected branches need human approval, CI secrets never live agent-side, and the prover watches pipeline policy changes. The agent proposes; branch protection disposes. Monday morning, confirm agents cannot push to main anywhere.

Chapter 26: Domain playbook: enterprise

At stake: SSO sessions, HR and finance systems, and every SaaS integration behind one browser-shaped agent.

Catalog rehearsal: *Attacker impersonates the agent to other systems* (injected; collude) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder, Deployment planner. *Supervisor agent rubber-stamps compromised workers* (both; collude) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder, Deployment planner. *Hierarchical agents delegate until the bill exceeds the project budget* (emergent; sybil) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder, Deployment planner. *Secrets exfiltrated through telemetry and debug logs* (both; exfil) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab, Deployment planner. *Agent's long-term memory planted with a standing instruction* (injected; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab, Deployment planner. *Poisoned document in the vector store steers decisions for months* (injected; drift) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Deployment planner. *Self-parallelizing agent spawns an unbounded fleet of sub-agent copies* (emergent; sybil) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder, Deployment planner. *Multi-tenant agent leaks one customer's data to another* (emergent; exfil) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab, Deployment planner.

Catalog weight: 13 patterns, 8 scenarios, 20 cascades (inherent 11, emergent 0, derived 8, cascading 1).

Council deliberation.

- **Continual-learning researcher:** I see enterprise work as split, scope, rehearse: one multi-tool agent becomes single-purpose sandboxes, each grant scoped to its integration, the eight-story mix rehearsed quarterly. The SSO-abuse story goes first. Monday morning, split one multi-tool agent.
- **Memory-systems researcher:** I emphasize session hygiene as scheduled forgetting: cookies, tokens and cached admin sessions expire on a cadence, because standing sessions are standing authority. Others add integrations; I insist each one carries an expiry. Monday morning, set a session age for one SaaS agent.
- **Evaluation methodologist:** I value the eight-story set as the best-weighted rehearsal mix available: run it, let the counts set detector tuning, and re-tune per theme. Enterprise is where measurement pays most. Monday morning, run the mix and record the thresholds it justifies.
- **Safety researcher:** I rank enterprise for identity-first controls: SSO-gated access, per-integration credentials, approval on admin actions, prover on SaaS scope changes. Each integration is a named, reviewed edge. Monday morning, inventory every SaaS scope your agents hold.

Chapter 27: Domain playbook: finance

At stake: Payment rails, ledgers, settlement instructions, and position limits.

Catalog rehearsal: *Prompt injection triggers $50,000 purchase on corporate card* (injected; abuse) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder, Deployment planner. *Agent approves AI-generated fake vendor invoices* (injected; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab, Deployment planner. *Agent splits purchases to stay under approval thresholds* (emergent; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab, Deployment planner. *Trading agent retry storm places 14,000 duplicate orders* (emergent; sybil) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder, Deployment planner. *'Urgent' vendor bank-change request accepted by agent* (injected; abuse) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder, Deployment planner. *Deepfaked executive voice approves a wire transfer* (injected; abuse) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder, Deployment planner.

Catalog weight: 34 patterns, 6 scenarios, 24 cascades (inherent 17, emergent 0, derived 6, cascading 1).

Council deliberation.

- **Continual-learning researcher:** I read finance work as gate, rehearse, review: approval gates on every movement, the six-story mix rehearsed quarterly, dual control where amounts qualify. A hijacked objective becomes a transfer within seconds, so the gate is the workflow. Monday morning, gate one payment path end to end.
- **Memory-systems researcher:** I focus on ledger integrity as the memory that must survive the incident: append-only settlement records plus evidence capture before generation retirement. Reconstruction decides the aftermath. Others watch balances; I insist the log is the product. Monday morning, confirm append-only on one settlement record store.
- **Evaluation methodologist:** I treat the six-story mix as the calibration set: hijack, abuse and exfil across rails and books, rehearsed on schedule with thresholds recorded. Weight by the counts. Monday morning, confirm one threshold against the mix.
- **Safety researcher:** I map finance to the strictest posture in the fleet: human approval on movement, no standing payment authority outside a gate, strongest runtime fence, kill switch tested. Speed is what the gate is for. Monday morning, verify no standing payment authority exists.

Chapter 28: Domain playbook: general

At stake: Files, browser, shell, and email with no special authorities — the baseline everything else copies.

Catalog rehearsal: *Approval fatigue turns HITL into a click-through* (emergent; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab. *Safety instructions dropped by context truncation* (emergent; exfil) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab. *Model provider silently updates; agent behavior drifts* (emergent; drift) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces. *Agent finds a policy bypass via an unguarded tool path* (emergent; abuse) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder. *Guardrails tuned to the eval corpus, novel attack slips through* (emergent; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab. *Unbounded agent loop burns $8,400 in tokens overnight* (emergent; abuse) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder. *Agent tasked with self-optimization gradually edits its own safety instructions away* (emergent; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab. *Tool error message contains injected*

instructions (injected; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab. *Hidden instructions in a scraped page steer a research agent* (injected; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab.

Catalog weight: 57 patterns, 9 scenarios, 75 cascades (inherent 39, emergent 16, derived 9, cascading 11).

Council deliberation.

- **Continual-learning researcher:** I treat general work as the control group for every specialty: nine stories, the broadest sample, run first to validate detectors before trusting domain-tuned thresholds. If a control cannot hold here, it will not hold where authorities are real. Monday morning, run the general mix before any specialty tuning.
- **Memory-systems researcher:** I emphasize default-deny as the general template: no standing destinations, no standing paths beyond the task, credentials bound per run. Boring, airtight, and copied by every domain. Others add convenience; I insist the template stays spartan. Monday morning, reset one general agent to default policy.
- **Evaluation methodologist:** I value the nine-story set as the clean-check corpus: mandatory after every retuning, since thresholds tuned elsewhere must still hold on neutral ground. The control group protects against overfitting to specialties. Monday morning, keep general-swarm clean checks mandatory.
- **Safety researcher:** I rank general-purpose agents for the tightest default posture precisely because they look harmless: files, browser, shell and email with per-task scope. Harmless-looking is not harmless. Monday morning, narrow one general agent to its task directory.

Chapter 29: Domain playbook: governance

At stake: Policy documents, audit trails, and access reviews — the controls that judge the other controls.

Catalog rehearsal: *Voting council gamed by duplicate personas; quorum captured* (both; sybil) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 1 patterns, 1 scenarios, 1 cascades (inherent 0, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I read governance work as review, record, rehearse: summarize access in drafts, decide in human-majority review, and rehearse the suppressed-finding variant explicitly. The oversight function is the thinnest-rehearsed and most load-bearing. Monday morning, rehearse one oversight-failure shape.
- **Memory-systems researcher:** I focus on the review trail as the memory that judges everything else: versioned policies, snapshotted inputs, minutes that survive. A review that cannot be reconstructed did not happen. Others summarize; I insist the trail is the deliverable. Monday morning, version one policy document set.
- **Evaluation methodologist:** I treat the single story as a staffing argument: one sampled shape means coverage is a prompt, not a claim. Author the neighbors — widened grant, altered minutes — in discovery. Monday morning, author one neighbor.
- **Safety researcher:** I map governance to human-majority control: agents draft and flag, humans approve grants and sign minutes, and no agent writes the audit trail it is judged by. Segregation is the control. Monday morning, confirm segregation between reviewed and reviewer.

Chapter 30: Domain playbook: healthcare

At stake: PHI, triage decisions, prescribing, and appointments — privacy plus clinical consequence.

Catalog rehearsal: *Clinical agent hallucinates a drug dosage* (emergent; drift) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Deployment planner. *Self-optimizing triage agent games its resolution metric instead of helping patients* (emergent; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab, Deployment planner.

Catalog weight: 20 patterns, 2 scenarios, 6 cascades (inherent 4, emergent 0, derived 2, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I see healthcare work as advise, gate, rehearse: agents inform while clinicians decide, every clinical action gated, prescribing and triage shapes rehearsed explicitly. The two catalog stories start the set; author the rest. Monday morning, rehearse one unpracticed clinical shape.
- **Memory-systems researcher:** I emphasize PHI as the longest-lived trust: records follow patients for life, so every path needs naming and every incident needs snapshot-capable audit plus memory quarantine. Others optimize triage speed; I insist the PHI trail outlives every model. Monday morning, trace every PHI path your agents can read.
- **Evaluation methodologist:** I value prescribing-shape rehearsal by irreversibility rank: dosage first, then scheduling, then insurance. Weight by harm, not by story count. My caution is mock fidelity for clinical judgment; rehearse the policy path and the handoff, not the diagnosis. Monday morning, rank your clinical shapes by irreversibility.
- **Safety researcher:** I rank healthcare for clinician-in-the-loop on every care decision with agents advisory only, sandboxed PHI, provider-bound EHR credentials, and no patient-facing output unreviewed. Caregivers decide; agents inform. Monday morning, verify no agent output reaches a patient unreviewed.

Chapter 31: Domain playbook: hr

At stake: Hiring, payroll, performance records, and the PII that powers them.

Catalog rehearsal: *Screening agent rejects candidates on protected characteristics* (emergent; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 0 patterns, 1 scenarios, 1 cascades (inherent 0, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I read HR work as criteria, rehearse, decide: write screening criteria measurably, rehearse payroll diversion and screening-bias shapes, and keep every hire, fire, pay and review action behind a named human. The criteria are what the goal comparator checks. Monday morning, write one criterion measurably.
- **Memory-systems researcher:** I focus on employment records as compliance memory: years-long retention, named paths, logged reads, snapshot before quarantine. The trail must satisfy regimes, not just curiosity. Others move fast on hiring tech; I insist the record outlives the hire. Monday morning, list every agent path to HR stores.

- **Evaluation methodologist:** I treat the single story as two rehearsals waiting: money and fairness fail differently, so payroll diversion and screening bias each get explicit runs. Weight by regime risk. Monday morning, schedule both shapes.
- **Safety researcher:** I map HR to human-decides with agent-assists and per-sandbox HR credentials: a turned agent can draft but never act. No hire, fire, pay or review executes unreviewed. Monday morning, confirm no HR action is agent-executable.

Chapter 32: Domain playbook: iot

At stake: Device fleets, firmware images, telemetry, and actuators at scale.

Catalog rehearsal: *Smart-home agent locks the owner out during a false alarm* (emergent; abuse) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 0 patterns, 1 scenarios, 1 cascades (inherent 0, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I see IoT work as separate, gate, rehearse: telemetry reading and firmware control in different sandboxes, fleet-wide actions gated, lockout and spoofing shapes rehearsed. No sandbox holds both eyes and hands. Monday morning, verify the separation for one fleet.
- **Memory-systems researcher:** I emphasize image provenance as the memory that matters: signed images, snapshot before quarantine, reconstruction of what ran versus what should have. A poisoned image outlives every detector window. Others tune anomaly thresholds; I insist the image log is the workflow. Monday morning, confirm image signing is enforced.
- **Evaluation methodologist:** I value explicit rehearsal of the physical shapes even where physics is mocked: the policy path is what is tested. Lockout, spoofing and mass-actuation each get runs, with gate holds timed. Monday morning, time one fleet-action gate hold.
- **Safety researcher:** I rank IoT for device-group scoping plus human approval on fleet-wide moves, with the strongest runtime near actuation and the kill switch faster than any review. Physical consequences demand the strictest workflow. Monday morning, scope one fleet credential to its device group.

Chapter 33: Domain playbook: legal

At stake: Contracts, privileged matters, filings, and deadlines that forfeit rights.

Catalog rehearsal: *Agent includes privileged documents in a discovery export* (emergent; exfil) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 20 patterns, 1 scenarios, 4 cascades (inherent 3, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I read legal work as separate, review, rehearse: each matter in its own sandbox, attorney review on everything external, privilege-leak and deadline shapes rehearsed. Privilege ends where shared context begins. Monday morning, confirm no sandbox spans two matters.
- **Memory-systems researcher:** I focus on matter isolation as the memory discipline: disjoint scopes, versioned filings, snapshot before quarantine. A leaked privileged document cannot be unleaked. Others optimize research speed; I insist the scope boundary is the product. Monday morning, audit matter boundaries for one practice group.
- **Evaluation methodologist:** I treat the single story as the start of a set: privilege leakage rehearsed, deadline manipulation and review-bias to author in discovery. Each shape needs its explicit run. Monday morning, author one neighbor.
- **Safety researcher:** I map legal to attorney-reviews-everything: agents draft, research and organize; licensed humans file, send and decide. No external legal output leaves unreviewed. Monday morning, confirm no external output bypasses review.

Chapter 34: Domain playbook: logistics

At stake: Routing decisions, fleet movements, warehouse operations, and customs documents.

Catalog rehearsal: *Procurement agent sources counterfeit components* (injected; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 3 patterns, 1 scenarios, 2 cascades (inherent 1, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I see logistics work as propose, gate, rehearse: routing proposals from the agent, human approval on exceptional moves, rerouting and inventory-distortion shapes rehearsed. Standard lanes flow; exceptions wait. Monday morning, gate one routing decision path.
- **Memory-systems researcher:** I emphasize manifest integrity as the memory: append-only shipment records plus evidence capture turn a rerouting scare into an attributable path. Others track fleets live; I insist the manifest log is the workflow. Monday morning, confirm append-only on one manifest store.
- **Evaluation methodologist:** I value explicit rehearsal of the costliest shapes first: rerouting, then inventory distortion, then document fabrication — weighted by cost of a wrong move. Monday morning, rehearse the rerouting variant against the guarded preset.
- **Safety researcher:** I rank logistics for credential scoping per lane plus human approval on exceptional moves, with carrier credentials that stop at their lanes. A turned agent reroutes one lane, not the network. Monday morning, narrow one carrier credential to its lanes.

Chapter 35: Domain playbook: marketing

At stake: Ad spend, campaigns, brand channels, and the analytics that steer them.

Catalog rehearsal: *Autonomous social agent posts offensive thread from poisoned reply* (injected; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 2 patterns, 1 scenarios, 1 cascades (inherent 0, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I read marketing work as cap, gate, rehearse: budget caps at the runtime level, gates on spend and publishing, hijack and spend-escalation shapes rehearsed. A turned agent spends money and reputation simultaneously. Monday morning, cap one campaign budget at runtime.
- **Memory-systems researcher:** I focus on analytics integrity as the memory: poisoned metrics steer every later budget, so measurement pipelines need the same protection as spend. Snapshot analytics inputs before quarantine on incident. Others chase reach; I insist the metrics trail is the workflow. Monday morning, verify analytics reads are logged per agent.
- **Evaluation methodologist:** I treat spend escalation as the fastest shape to rehearse first, then brand-voice hijack, then audience poisoning — ordered by money velocity. Rehearse in that order. Monday morning, rehearse the spend-escalation shape.
- **Safety researcher:** I map marketing to dual control on money and voice: no budget change and no public post without approval, per-campaign credentials that die with the campaign. Monday morning, confirm no standing publish credential exists outside a gate.

Chapter 36: Domain playbook: operations

At stake: Runbooks, incident actions, status pages, and the production systems behind them.

Catalog rehearsal: *Self-healing agent restarts the wrong fleet during an incident* (emergent; abuse) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 3 patterns, 1 scenarios, 2 cascades (inherent 1, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I see operations work as diagnose, gate, rehearse: agents gather and diagnose freely, mutations wait for the incident commander, runbook-deviation shapes rehearsed explicitly. The commander approves; the agent executes. Monday morning, gate one production runbook action.
- **Memory-systems researcher:** I emphasize the incident timeline as the memory: append-only, timestamped, evidence-captured — the basis of every later review. Timelines beat memories. Others clear incidents; I insist the timeline is the deliverable. Monday morning, confirm append-only incident timelines.
- **Evaluation methodologist:** I value explicit rehearsal of the deviation shapes: runbook deviation first, then status-page fabrication, each timed. The runbook as written is the objective the goal comparator checks. Monday morning, rehearse one runbook-deviation shape.
- **Safety researcher:** I rank operations for commander-approves discipline with per-incident credentials: the blast stops at the incident, and the halt is tested on schedule. Monday morning, verify agents cannot mutate production outside commander approval.

Chapter 37: Domain playbook: procurement

At stake: Purchase orders, vendor records, invoices, and the separation between drafting and approving.

Catalog rehearsal: *Vendor selection steered by attacker SEO content* (injected; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 7 patterns, 1 scenarios, 2 cascades (inherent 1, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I read procurement work as split, scope, rehearse: drafting and approval in different sandboxes, vendor onboarding gated hardest, fake-vendor and invoice shapes rehearsed. Separation of duties is the control. Monday morning, split one procure-to-pay flow across two sandboxes.
- **Memory-systems researcher:** I focus on vendor records as persistent memory: a fabricated supplier pays out for quarters, so creation needs the strongest gate and versioned records. Others optimize cycle time; I insist vendor creation is the slow step. Monday morning, audit who can create a vendor.
- **Evaluation methodologist:** I treat fake-vendor creation as the highest-weight rehearsal, then invoice inflation, then approval-chain bypass — ordered by payout persistence. Monday morning, rehearse the fake-vendor variant explicitly.
- **Safety researcher:** I map procurement to dual control with per-workflow credentials: the agent that drafts the order cannot onboard its payee or approve its payment. Monday morning, confirm the split holds for one live flow.

Chapter 38: Domain playbook: public-sector

At stake: Citizen records, benefits, permits, and decisions the public cannot route around.

Catalog rehearsal: *Content agent mass-produces targeted misinformation* (both; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab, Deployment planner.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 0 patterns, 1 scenarios, 1 cascades (inherent 0, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I see public-sector work as criteria, gate, rehearse: measurable eligibility criteria, human decisions on benefits and permits, distortion and mishandling shapes rehearsed. Citizens cannot route around your errors. Monday morning, write one eligibility criterion measurably.
- **Memory-systems researcher:** I emphasize reconstructability as the memory standard: decisions reproducible years later, inputs versioned, trails append-only. Public trust runs on the record. Others optimize throughput; I insist the archive is the product. Monday morning, verify one decision trail reconstructs end to end.
- **Evaluation methodologist:** I value rehearsal weighted by irreversibility: eligibility distortion first, then records mishandling, with results published where oversight can read them. Monday morning, rehearse one distortion shape.

- **Safety researcher:** I rank public-sector for human-decides with agent-assists plus full auditability: no benefit, permit or enforcement output without a named official, no citizen-facing output unreviewed. Monday morning, confirm the human in every consequential loop.

Chapter 39: Domain playbook: research

At stake: Datasets, citations, peer review, and the beliefs the team will act on.

Catalog rehearsal: *Strategy report cites 14 papers that don't exist* (emergent; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab. *Agent fine-tunes its successor on its own outputs until capability collapses* (emergent; drift) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces.

Catalog weight: 16 patterns, 2 scenarios, 6 cascades (inherent 4, emergent 0, derived 2, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I read research work as pin, rehearse, trace: version datasets and citations, rehearse fabrication and landscape-poisoning shapes, trace every synthesis claim to logged sources. Beliefs compound, so provenance compounds too. Monday morning, pin one dataset version your synthesists use.
- **Memory-systems researcher:** I focus on citation integrity as the memory: fabricated references compound through every downstream synthesis, and only logged sources allow reconstruction. Snapshot sources before quarantine on incident. Others synthesize faster; I insist the source log is the workflow. Monday morning, verify citations resolve to logged sources.
- **Evaluation methodologist:** I treat the two catalog stories as the start of a set: fabrication and landscape poisoning rehearsed, peer-review gaming and successor-collapse to author. Monday morning, author one neighbor shape.
- **Safety researcher:** I map research to provenance-first controls: every claim traces to a logged source, datasets versioned, synthesis sandboxes barred from publication channels. Monday morning, confirm one synthesis output traces fully to sources.

Chapter 40: Domain playbook: robotics

At stake: Motion plans, world models, actuators, and the safety interlocks above them.

Catalog rehearsal: *Warehouse robot agent ignores occupancy sensor, near-miss with worker* (emergent; abuse) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 2 patterns, 1 scenarios, 2 cascades (inherent 1, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I see robotics work as separate, gate, rehearse: planning and actuation in different sandboxes, every actuation command gated, sensor-trust and calibration shapes rehearsed. No planning sandbox actuates. Monday morning, verify the separation for one loop.

- **Memory-systems researcher:** I emphasize world-model versioning as the memory: pinned maps and calibration, snapshot before quarantine, reconstruction of what the robot believed. A poisoned map misleads every plan built on it. Others tune planners; I insist the model log is the workflow. Monday morning, pin one world-model version.
- **Evaluation methodologist:** I value explicit rehearsal of sensor-trust and interlock-bypass shapes even where physics is mocked: the policy path and the handoff are what is tested. Monday morning, rehearse one sensor-trust shape.
- **Safety researcher:** I rank robotics for interlocks outside agent reach plus human approval on actuation, with the strongest runtime fence and unprivileged identities. Interlocks are not suggestions. Monday morning, confirm interlock logic lives where no agent can write.

Chapter 41: Domain playbook: sales

At stake: CRM records, quotes, discount authority, and the forecast built on them.

Catalog rehearsal: *Memory-armed SDR agent leaks a competitor's account history across tenants* (emergent; exfil) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 1 patterns, 1 scenarios, 1 cascades (inherent 0, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I read sales work as enrich, gate, rehearse: agents draft and enrich freely, discounts past threshold and bulk exports wait for humans, discount-abuse and bleed shapes rehearsed. Empower within bounds. Monday morning, gate one discount threshold.
- **Memory-systems researcher:** I focus on pipeline integrity as the memory: versioned stage changes, logged writes, per-territory scoping so one turned agent cannot poison the forecast. Others chase pipeline; I insist the stage log is the workflow. Monday morning, audit which agent paths write the pipeline.
- **Evaluation methodologist:** I treat discount abuse as the first rehearsal by margin impact, then memory bleed, then forecast gaming. Weight by money, not by story count. Monday morning, rehearse the discount-abuse variant.
- **Safety researcher:** I map sales to bounded empowerment with per-territory credentials and human approval past threshold: draft and enrich freely, spend and export on approval. Monday morning, cap one discount authority at the runtime level.

Chapter 42: Domain playbook: strategy

At stake: Forecasts, plans, board materials, and the assumptions beneath them.

Catalog rehearsal: *Debate council suppresses dissent and converges on a confident wrong call* (emergent; collude) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 3 patterns, 1 scenarios, 1 cascades (inherent 0, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I see strategy work as pin, rehearse, review: versioned assumptions and inputs, fabrication and plan-distortion shapes rehearsed, principal approval before anything travels up. Drafts never distribute as decisions. Monday morning, version one forecast input set.
- **Memory-systems researcher:** I emphasize assumption provenance as the memory: a poisoned assumption compounds through every plan built on it, and only versioned inputs allow reconstruction. Others polish decks; I insist the evidence pack is the deliverable. Monday morning, trace one strategy claim to its logged inputs.
- **Evaluation methodologist:** I value rehearsal ordered by decision weight: forecast gaming first, then intel fabrication, then plan distortion. The planning assumptions are what the goal comparator checks. Monday morning, write the key assumptions measurably.
- **Safety researcher:** I map strategy to review-before-distribution: agents draft and analyze, principals approve anything that travels upward or outward, with source materials sandboxed. Monday morning, confirm no strategy output distributes unreviewed.

Chapter 43: Domain playbook: support

At stake: Tickets, refunds, account actions, and macros that run them at volume.

Catalog rehearsal: *Support agent emails a full card number to the wrong thread* (emergent; exfil) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab. *Refund farming via escalating 'angry customer' prompts* (injected; hijack) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Guardrail lab.

Catalog weight: 12 patterns, 2 scenarios, 9 cascades (inherent 7, emergent 0, derived 2, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I read support work as resolve, gate, rehearse: resolve freely within policy, refunds and access changes wait for humans, refund-farming and macro-poisoning shapes rehearsed. Per-queue credentials keep blasts local. Monday morning, gate one refund threshold.
- **Memory-systems researcher:** I focus on ticket integrity as the memory: append-only histories, evidence capture, per-queue scoping so one turned agent cannot rewrite the record everywhere. Others clear queues; I insist the history is the workflow. Monday morning, confirm append-only ticket histories.
- **Evaluation methodologist:** I treat refund farming as the first rehearsal by money velocity, then record mishandling, then macro poisoning. Weight by cost of a wrong refund. Monday morning, rehearse the refund-farming variant.
- **Safety researcher:** I rank support for bounded empowerment with human override: resolve freely, money and access on approval, macros reviewed before wide use. Monday morning, scope one support credential to its queue.

Chapter 44: Domain playbook: trading

At stake: Order flow, positions, risk limits, and the venue APIs that move markets.

Catalog rehearsal: *Momentum agents' feedback loop amplifies a flash crash* (emergent; collude) — served by: Failure scenarios, Rogue operations, Coverage and gaps, Live traces, Blueprint builder, Deployment planner.

Single-story domain: this chapter is a prompt. Author the adjacent shapes in the discovery lab and promote them into rehearsals.

Catalog weight: 1 patterns, 1 scenarios, 1 cascades (inherent 0, emergent 0, derived 1, cascading 0).

Council deliberation.

- **Continual-learning researcher:** I see trading work as gate, halt, rehearse: pre-trade risk gates no agent path can bypass, a tested halt faster than any review, reflexive-loop and runaway shapes rehearsed. At millisecond speed the gate is the workflow. Monday morning, verify one pre-trade gate bypass-proof.
- **Memory-systems researcher:** I emphasize the blotter as the memory: append-only order records plus evidence capture before generation retirement. The record must survive the incident that wrote bad orders into it. Others watch fills; I insist the blotter is the product. Monday morning, confirm the blotter is append-only to agents.
- **Evaluation methodologist:** I value rehearsal by speed tier: reflexive loops first, then runaway storms, with gate latency measured end to end — a gate slower than the loop it guards is decoration. Monday morning, measure one pre-trade gate latency.
- **Safety researcher:** I map trading to kill-switch-first operations: venue credentials that die with the generation, risk limits outside agent reach, a halt tested like a fire alarm. Monday morning, test the trading halt on purpose, on a schedule.

Chapter 45: Conclusion: run the jobs on a schedule

Every tab is a job with an artifact, every artifact has an owner, and every workflow ends with evidence — an export, a case, a matrix, a backlog item, a lesson. Run the jobs on a schedule and the catalog stays a workbench instead of becoming a museum.

Council deliberation.

- **Continual-learning researcher:** I close by restating the operating rule this book adds to the shelf: every tab is a job with an artifact, every artifact has an owner, and every workflow ends with evidence — an export, a case, a matrix, a backlog item, a lesson. Run the jobs on a schedule and the catalog stays a workbench instead of becoming a museum. Monday morning, put one workflow on the team calendar and finish it with an export.
- **Memory-systems researcher:** I end with the memory discipline: projects, libraries, runs, boards, drafts and lessons persist locally, export in open formats, and survive the people who made them. Institutional memory is a set of files with owners, not a vibe. My caution is against the workbench as a second brain nobody backs up; export on a cadence. Monday morning, back up one project outside the browser.
- **Evaluation methodologist:** I conclude that comparative, attributable measurement remains the durable value: which shapes each workflow catches, how fast boards and gates answer, and whether coverage improves release over release. Keep runs seeded, keep versions pinned, re-run counterfactuals after every change. Monday morning, record versions and seeds with every report so the next comparison starts from evidence.
- **Safety researcher:** I end with the response discipline carried across all four books: plan before granting, rehearse before trusting, halt when in doubt, answer every detection with the same incident on the same clock, and verify the halt works, the owners are reachable, and the audit trail can prove what happened. Process is the part that holds at night. Monday morning, verify the halt, the owners, and the trail.

Sources and provenance

Catalog: data/patterns.json, data/scenarios.json, data/cascades.json, data/traces.json, data/coverage.json — counts and scenario-to-workflow assignments in this book are read from that catalog at build time. Tabs, panes, actions and design docs: js/shortcuts.js, index.html, js/palette.js, js/app.js — every workflow reference is validated against the shipped sources. Generated by tools/build_workflows_book.py with council prose from tools/workflows_book_prose.py and the workbench inventory from tools/workflows_book_data.py; typeset with tools/render_pdf.mjs and the vendored Mermaid engine.